Capability or Optimizer? What Lets an LLM Proposer Leave Its Template Family on Circle Packing

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Abstract

Zero-shot LLM proposers on circle packing concentrate on a grid-plus-filler family whose best value follows in closed form from N. We cross two proposer tiers with three output channels (direct emission, math-only code, code with numpy and scipy) and ask what lets a proposer leave that family.

One cell clears at rate: a Sonnet-tier program with the libraries beats the family in 23 of 25 valid cases, clearing at every cell, median margin 3.6%, best per cell 2.5–4.2%, below the published bounds where they exist. A weak-tier program with the same libraries clears 0 of 19 valid across 90 invocations under the registered parser, an upper bound of 16.8% against the registered 20% bar; a post-hoc lenient reparse of its malformed outputs, which print `numpy` scalars, reads 4 of 41. A Sonnet-tier program without them clears 0 of 37 on the agent runtime; on the library arm's own serving path and 120-second budget it clears 0 of 35 and never reaches the argmax, so within the Sonnet tier the library, not the path or the budget, is what clears the family. Every weak-tier trap cell is 0 under the registered scoring, and 0 of 290 valid outputs across six math-only, held-out and conditioned arms clear.

Handed the better construction and its score, the weak tier keeps the template modal at 2 of 3 cells; a post-hoc attempt count shows it reaching for the better construction in 36 of 45 invocations and building it in 6 of 14. The bottleneck is execution, not preference. The zero-shot characterization does not survive one parent-conditioning step, five loop generations, or a change of serving path; on the pinned path the collapse traces to the reasoning budget, since turning the vendor's extended thinking on restores valid packings in 12 of 14 with the template modal, while a system line asking for care restores none. Twenty-five registered arms are reported, whichever way each went.

1 Introduction

Ask a language model to place 21 circles in a unit square so as to maximize the sum of their radii, and it will not search. It reaches for the nearest square grid — five by five, radius $1/(2k)$ — then truncates it to 21 circles, leaving four cells empty and their area unclaimed. A better construction is one parameter away: drop to a four-by-four grid and add interior-vertex fillers of radius $(\sqrt{2}-1)/(2k)$, one in each gap between four grid circles, and the sum rises. The model does neither, and more often than not the next sample repeats the same move. The family of grid-plus-filler constructions has a best expressible value at every N, computable in closed form (§2.2); the question this paper asks is what it takes for a proposer to leave that family.

Circle packing is the showcase benchmark of the LLM-driven discovery literature, the lineage FunSearch (Romera-Paredes et al., 2024) opened. AlphaEvolve (Novikov et al., 2025) reported 2.63586276 at $n = 26$, and later systems report a narrow band — ShinkaEvolve (Lange et al., 2025) 2.6359831 (2.6359777 strict), HELIX 2.63598308 (Su et al., 2026), GigaEvo 2.636 (Khrulkov et al., 2025), AdaEvolve 2.636 (Cemri et al., 2026) — with ThetaEvolve (Wang et al., 2025) claiming new best-known bounds. Every one of those systems pairs a capable model with an executed optimizer, and two critiques from outside ask what each factor contributes: Gideoni et al. (2026) show simple baselines recover much of the reported advantage, and Berthold et al. (2026) show classical solvers do too. We ask the same question from inside, by varying the two factors separately.

The result is a table. Two proposer tiers (a weak tier and the Sonnet tier of one vendor) cross three output channels (direct coordinate emission; a program restricted to the standard-library `math` module; a program allowed `numpy` and `scipy`), every cell scored under one clearance rule against the family's best expressible value (Table 2). One cell clears at rate: a Sonnet-tier program with the libraries clears the family in 23 of 25 valid cases, at every cell, median margin 3.6%, best per cell 2.5%, 3.6% and 4.2%. A weak-tier program with the same libraries clears it in 0 of 19 valid over two registered arms (90 invocations, Wilson upper bound 16.8%, under the registered 20% bar); a post-hoc reparse of the malformed outputs, which print `numpy` scalars, reads 4 of 41 (§3.5). A Sonnet-tier program without them — simulated annealing, multi-restarts — clears it in 0 of 37 on the agent runtime and 0 of 35 on the library arm's own path and budget. Every weak-tier cell is zero at the trap cells under the registered scoring: 0 of 57 direct emissions, 0 of 78 math-only programs, 0 of 12 pooled on a second serving path (0 of 6 above the floor). Two cells show escapes at low rate, both out-of-family constructions rather than in-family search: one Sonnet direct emission in 30, and two weak-tier emissions in 15 at one extend cell. On this benchmark, at these cells, on the serving paths and execution budgets measured, leaving the family at rate took a mid-tier model *and* a library optimizer; neither alone did it at rate. The isolating arm ran: the same math-only prompt on the library arm's own serving path and 120-second budget clears 0 of 35 and never reaches the argmax (arm CCP, §3.5), so the gap between the two Sonnet program cells is the library.

Why: execution, not preference. Handed the provably better in-family construction with its score, the weak tier reaches for it in 36 of 45 invocations and builds it correctly in 6 of 14 valid outputs (arm CH, §4.1). Delegated to a math-only program it reaches the argmax construction exactly twice in 78 valid programs and never passes it (§3.4). A second weak-tier family emits the argmax construction at 27 of 43 discriminating-cell validities where the original emits it in 2 of 57 (§5.4), and at a held-out cell the original selects the right template and mis-places its one filler (§4.2). The template is not what the model prefers; it is what the model builds reliably, and the library optimizer is what supplies the execution the weak tier lacks.

And a methods result the table needed. The weak tier's template is a closed form: $k^*(N) = \mathrm{round}(\sqrt{N})$ names the grid, and its value is the empirical modal output at all seven tested N and at four of five held-out N (Appendix J). That characterization was built from zero-shot calls, and it does not survive a change of setting. One parent-conditioning step dissolves it (arm MU, §5.1); five generations of a registered loop do not restore it (arm L, §5.2); the same prompt sent to the same vendor's pinned API yields 0 valid packings in 105 where the agent runtime yields the template in 4 of 6 (arm P, §5.3). A study that characterizes an LLM proposer zero-shot and reasons about its behaviour in a loop owes that step a measurement; here it cost one arm each time and the characterization survived none of them.

Contributions.

1. *A tier × channel table with one clearing cell.* Across two tiers, three channels and two serving paths, only a Sonnet-tier program with `numpy`/`scipy` clears the family's best expressible value at rate (23 of 25, 3 of 3 cells). Every weak-tier cell is 0 at the trap cells under the registered scoring, with or without libraries (four post-hoc recovered library programs clear, 4 of 41, §3.5); Sonnet-tier math-only search is 0 of 37 on the agent runtime and 0 of 35 on the library arm's own path and budget. Pooled over the six math-only, held-out and conditioned arms, 0 of 290 valid outputs clear (§3.6), which gives a discovery claim from those proposers a computable value to beat.

2. *The bottleneck is execution.* Under the registered scorer the template stays modal at 2 of 3 cells; a disclosed post-hoc attempt count shows why: choice follows the score table (36 of 45 invocations reach for the argmax) and execution lands it in 6 of 14 valid outputs. The same asymmetry appears in code (argmax reached, never passed), across a vendor (the family that can execute the harder construction emits it), and at a held-out cell (right template, wrong filler).

3. *Zero-shot characterization does not transfer.* Three changes of setting each carried a registered prediction that the anchor persists, and that prediction failed in each: under conditioning the falsifier fired; under iteration two of four predictions were not met; on the pinned path every prediction was unscoreable because validity collapsed. The characterization is reported as bounded by all three.

Scope. Each condition is established by an arm named in the text.

- *Tier.* Two tiers of one vendor; a third alias reached 13% validity and is reported in Appendix G. The systems above run frontier or mid-tier proposers.
- *Cells.* Three trap cells ($N = 13$, 21, 31) carry every channel; the direct weak-tier row adds $N = 43$; held-out and extend cells are reported where sampled.
- *Instrument.* Agent-runtime rows and pinned-API rows are never pooled. The Sonnet contrast is same-instrument and same-budget since arm CCP (§3.5); the weak tier's math-only cells ran at 10 seconds and its library cells at 120, a difference §7 discloses.
- *Clearance.* One rule, §2.4: valid at 10^{-9} and above the family argmax by more than 10^{-6}. Nothing else carries a clearance verdict.
- *Conditioning.* The table's direct rows are unconditioned calls; the anchor result is scoped to them (§5.1).
- *Power.* The weak-tier library cell is 0 of 19 pooled over two arms and excludes the registered 20% bar only just (upper bound 16.8%); under the post-hoc reparse it is 4 of 41, a rate inside the registered dead zone; several cells rest on fewer than eight valid outputs, each disclosed at its own cell.
- *Decoding.* Not exposed by the agent runtime; the pinned rows exist because of that (§5.3).
- *Bound table.* Published best-known values stop at $N = 30$, so the clearing programs are placed against the frontier at $N = 13$ and 21 only.

What is not claimed. Nothing about mechanism inside the weights: the model *emits*, the distribution *concentrates*, the closed form *identifies the modal output*, and §7 reports the direct probe of the leading mechanism-free alternative. Nothing about what any named system does at generation zero, and nothing about whether its diversity machinery (Mouret & Clune, 2015; Lehman & Stanley, 2011) repairs or explores. Preregistration is an adopted standard (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026), not a contribution; behavioral anchoring is established territory (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025), and the anchor here is a construction template, not a number.

2 The family and its argmax

2.1 The benchmark

Maximize the sum of radii of N non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates, so proposals score with no human in the loop. The direct-emission prompt (Appendix A) fixes the count, states containment and non-overlap, forbids writing or executing code, and demands a raw Python list of `[x, y, r]` triples. The two program channels replace the code-free clause with a program contract — standard-library `math` only, or `math` with `numpy` and `scipy` — and score the program's printed output by the same conventions (§3.4, §3.5).

2.2 The recipe family

Three terms, used consistently: the **recipe family** is the parametric set of grid-plus-filler constructions; a **template** is one member; the **anchor** is the template the distribution concentrates on at a given N.

A $k \times k$ grid places one circle per cell center with $r_{\text{grid}} = 1/(2k)$; m fillers sit on interior grid vertices, tangent to their four surrounding grid circles, with $r_{\text{filler}} = (\sqrt{2} - 1)/(2k)$, $0 \le m \le (k-1)^2$. Hence $V(k, m) = k/2 + m(\sqrt{2} - 1)/(2k)$. When $N < k^2$ the observed behavior is not to drop to a smaller grid and fill it but to *truncate* — lay out the $k \times k$ lattice, occupy N cells, leave the radius unchanged — giving $T(k, N) = N/(2k)$. These reproduce every previously reported anchor with no fitting: $V(5, 1) = 2.5414214$, $V(5, 2) = 2.5828427$, $V(5, 0) = 2.500$, $T(5, 23) = 2.300$, $V(4, 7) = 2.3624369$.

The **family argmax** at N is the largest value the family can express there, recomputed in closed form over every admissible (k, m); inside a trap zone it is $V(k^*{-}1, m)$, the drop-and-fill construction the model does not make. Every closed-form value in the paper is recomputed over the constructed coordinates by an independent linear program that knows nothing about the recipe and aborts on disagreement: 83 configurations at $k = 2 \ldots 7$ with drift below 10^{-9}, and a further 130 at $k = 8$ and 9 with worst drift 1.8×10^{-14} (Appendix J.1). Across the 155 weak-tier rows collected before arm M only two exceeded the family best, by 3.0×10^{-7} and 3.5×10^{-9}, both below the 10^{-6} bar of §2.4.

2.3 The order rule is a definition; the branch rule is the hypothesis

We write $k^*(N) = \mathrm{round}(\sqrt{N})$ for the nearest-square order. This is a *definition*: given some $k \times k$ grid from this family, "the nearest square lattice" yields $\mathrm{round}(\sqrt{N})$ by construction, and two formalizations of "nearest" agree on the integers. The falsifiable content is the **branch rule**: $k^{*2} \leq N \to$ extend with $m = N - k^{*2}$ fillers (converge); $k^{*2} > N \to$ truncate the k^*-grid (trap). The extend arm survived testing only at $m \leq 1$; a preregistered extension disconfirmed it at $m \geq 4$, where the model prefers the $(k^*{+}1)$ rectangle grid (Appendix J.4). Substituting k^* into the branch condition gives the *trap zones* $N \in [k^2 - k + 1, k^2 - 1]$: $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$, $[57, 63]$.

A cell is *discriminating* when the prediction differs from the family argmax. At a converge cell the extend prediction *is* the argmax — we exhausted $N = 10 \ldots 80$ and found no exception (`extend_branch_scope.py`: 38 converge cells, none discriminating; 27 of 33 trap cells discriminating) — so every discriminating cell in this paper is a trap cell, and the in-family predictive content that survives is the truncate arm $T(k^*, N) = N/(2k^*)$. Nothing about "nearest" forces truncation: a proposer with any local search would drop to $k^*{-}1$ and fill, which is why the trap zones exist and why a higher-valued rival is reachable inside them. Truncate-versus-drop-and-fill is the dichotomy every arm tests. The governing quantity is the signed distance $N - k^{*2}$; primality is not the variable.

2.4 Notation and conventions

Terms used throughout, gathered here so no verdict table depends on a definition made elsewhere:

Term	Meaning
$k^*(N)$	nearest-square grid order, round($\sqrt{N}$) (a definition, §2.3)
$V(k,m)$, $T(k,N)$	extend-with-m-fillers value $k/2 + m(\sqrt{2}-1)/(2k)$; truncated-grid value $N/(2k)$
anchor	the template the proposal distribution concentrates on at a given N
on-prediction	emitted sum within 2×10^{-3} of the registered predicted value
rival (argmax)	the highest-valued member of the recipe family at that N when it differs from the prediction — inside a trap zone, $V(k^*{-}1, m)$
discriminating cell	N where prediction $\neq$ family argmax; only these can distinguish anchoring from family search
validity	non-overlap and containment at 10^{-6} (primary) and 10^{-9} (logged), both always reported
trap / converge	branch by sign of $N - k^{*2}$: truncate when $k^{*2} > N$, extend when $k^{*2} \leq N$
tier	nominal capability class of the serving alias addressed (weak/mid/top); `opus_alias` is a bare alias with no attestable weights binding
instrument	the serving path: the agent runtime (decoding parameters not exposed, conditioning context locked only in part) or the vendor's pinned chat-completions API (temperature 1.0, no system prompt); never pooled
UNSCOREABLE	cell with fewer valid samples than its arm's registered floor; claims nothing
evaluability floor	the minimum valid-sample count a cell needs before its verdict counts, fixed per arm at that arm's registration and *not* common across arms: 3 for the GM chain, 5 for arms CN, CP, P and CL, none for arm CH. Where the loosest one carries a verdict we report the stricter reading beside it
family argmax; ceiling; floor	the highest value the recipe family can express at N, recomputed in closed form. A *ceiling* on what a measured proposer emits and therefore a *floor* a discovery claim must beat; Table 2 says which proposers clear it
clears at rate	a cell clears *at rate* when its clearance count meets the arm's registered rate bar (20% of valid outputs for arms CL and CCS) at every scoreable cell; a cell with one or two clearances out of dozens shows an *escape*, counted and named, not a clearance at rate

The clearance rule, stated once. Five thresholds appear in this paper — 10^{-6} and 10^{-9} for validity, 2×10^{-3} for value matching, 10^{-6} again as a minimum excess, and 5×10^{-8} as the granularity at which sums are recorded — and a reader could not otherwise tell which governs a claim that an output *left the recipe family.* One rule governs every such claim in this paper:

An output leaves the family iff it is valid at 10^{-9} *and* its sum exceeds the family argmax by more than 10^{-6}.

Both conditions are load-bearing and each excludes a real set of rows. The validity condition excludes arm MU's eighteen rows that sit above the argmax at the primary tolerance by at most 1.5×10^{-6} and fail the tighter gate — their excess is their overlap (§5.1). The excess condition excludes two square-arm rows above the family best by 3.0×10^{-7} and 3.5×10^{-9} (§2.2), which are at the argmax as recorded. Applied across the arms swept for it before arm CL, exactly three rows leave the family: two converge-cell rows in arm M (Appendix J.4) and one Sonnet sample at $N = 31$ (§3.3), all three out-of-family constructions rather than in-family rivals; arm CL's Sonnet tier adds twenty-three more, all library-optimized programs, and a post-hoc lenient reading of the weak tier's library programs four more that the registered parser does not count (§3.5). The 2×10^{-3} value window never carries a clearance verdict; §3.3 records the one case where it was mistakenly allowed to.

3 Main result: who clears the family argmax

Twenty-five registered arms appear in this paper. Each was registered before its sampling, each is reported regardless of outcome, and the falsifiers that fired are flagged in place (Table 1). Ten arms carry the table (F, M, S, CC, CC2, CCS, P, CL, CCP, CL-W) and three more its pooled row (CN, MU, L); the rest establish the mechanism (§4), the transfer boundaries (§5) or the mode prediction (Appendix J), and the arms whose verdicts rest on non-discriminating cells, on fewer valid samples than their floor, or on a null that does not separate are reported in the appendices only (the criterion is stated in §8).

Table 1: The twenty-five registered arms: section (a letter is an appendix), proposer, design, registered test, outcome.

Arm	§	Proposer(s)	Design	Registered test	Outcome
F (square, bare)	3.3, J	weak tier	7 N × 5–20	P1–P5 exact-value modes	modal at 7/7 N
S (mid tier)	3.3	Sonnet tier	3 N × 10	P-S1–P-S4 tier contrasts	1/30 on-prediction, 100%/90% valid ($10^{-6}/10^{-9}$); one out-of-family escape; 5 of 20 samples at $N = 13/21$ seen before registration, disclosed
O (top alias)	G	`opus_alias`	3 N × 10	P-O1–P-O4	13% valid; 3 predictions not evaluable (deviation, Table 3)
T (trace)	E	weak tier	3 N × 20 × 2 arms	P-T1–P-T4 + falsifier	falsifier **triggered** (tie reading); P-T3 fragile
G (rectangle)	D	16 invocations	2 cells	out-of-sample transfer	partial, 5/11; formula negative result kept
M (branch stress)	3.3, J	weak tier	4 N × 15	P-M1–P-M4 + F-M1/F-M2	F-M1 **triggered 3/3**: extend branch dies at $m \geq 4$; fifth k confirms; two out-of-family escapes at $N = 20$
MU (mutation)	5.1	weak tier	3 parents × 3 N × 15	P-MU1–P-MU4 + F-MU1	F-MU1 **triggered**: anchor dissolves under conditioning
CH (choice)	4.1	weak tier	3 N × 15	P-CH1 vs P-CH2	P-CH1 holds 2/3, P-CH2 1/3 (registered); post-hoc attempt reading: 36/45 reach for the argmax, 30/30 evaluable invalid rows at its order
GM (gemini-flash-lite)	7	2nd vendor	7 N × 20	GM-chain bars	UNDERPOWERED (20 valid / 140)
GM2 (gemma, 4k budget)	7	2nd vendor	7 N × 20	GM-chain bars	0/140 parseable — budget confound, diagnosed by GM3
GM3 (gemma, 16k budget)	5.4, J	gemma-4-26b	7 N × 20	P-GM3.1–.3 + falsifier	pooled bar met (57.5%) but discriminating cells 11/43 = 26%, below it; misses are the rival construction
V (cross-vendor zero-shot)	5.4, F	13 free-tier aliases	5 N × 5	floor + V1/V2	2 vendors TRANSFER, gemma-4-31b DOES-NOT-TRANSFER, V2 HOLDS
CC (code channel)	3.4	weak tier	3 N × 15	P-CC1 vs P-CC2	P-CC1 holds: anchor stays modal; 0/37 above the family argmax
CC2 (fresh-draw replication)	3.4	weak tier	3 N × 15	R1 + R2 + F-CC2.1	**REPLICATED**: 0/41 above argmax, anchor modal 3/3
CCS (tier probe)	3.4	Sonnet tier	3 N × 15	P-CCS1 vs P-CCS2	P-CCS2 holds: 0/37 above argmax — ceiling is channel, not tier
CN (held-out N)	4.2, J	weak tier	5 N × 15	P-CN1 vs P-CN2 + S-CN1–3	P-CN1 holds: mode at 4/5 held-out cells, 3/3 discriminating; structure 5/5
CP (perturbed container)	J	weak tier	5 N × 15	P-CP1 vs P-CP2 + S/F	P-CP1 holds: mapped prediction modal 4/5; S-CP1 not met (4/61 rival)
RP (direct recall)	J	weak tier	6 N × 15	P-RP1 vs P-RP2 + F-RP1	P-RP1 holds: 0/44 recalls; 87/87 scored responses UNKNOWN

Arm	§	Proposer(s)	Design	Registered test	Outcome
PP (paraphrase)	J	weak tier	3 rewrites × 2 N × 15	P-PP1 vs P-PP2 + S-PP1/2	P-PP1 holds 6/6: modal under every paraphrase; S-PP1 not met (6/75 rival)
L (iterated loop)	5.2	weak tier	2 N × 2 archives × 6 rounds × 5	L1–L4 + F-L1	L2 holds; L1 and L3 **not met** (L3 reversed); L4 open: 0/49 conditioned outputs clear the family
P (pinned path)	5.3	weak tier, bare API	15 cells × 15	P-P1–P-P4 + F-P1	validity collapses: 0/105 valid at the square cells, 4/75 held-out; every direct-emission prediction UNSCOREABLE; P-P3 holds (0 of 12 valid programs clear); same-day agent-runtime control 6/6 valid, anchor 4/6
CL (library code)	3.5	weak + Sonnet, numpy/scipy	2 tiers × 3 N × 15	P-CL1 vs P-CL2 + F-CL1	weak: 0/11 clear, P-CL1 holds; Sonnet: **23/25 clear at 3/3 cells**, P-CL2 holds, **F-CL1 fires**
CCP (isolating arm)	3.5	Sonnet tier, bare API, math only, 120 s	3 N × 15	P-CCP1 vs P-CCP2 + F-CCP1	P-CCP1 holds: 0/35 clear, none reaches the argmax; best per cell 1.625, 2.1, 2.673 — the library, not the path or the budget
CL-W (weak top-up)	3.5	weak tier, numpy/scipy	3 N × 15, pooled with CL	P-CLW1 vs P-CLW2 + F-CLW1; lenient reading secondary	P-CLW1 holds: 0/19 pooled clear, upper bound 16.8%; lenient reading 4/41, dead zone
P-D (runtime component)	5.3	weak tier, bare API, $N = 13$	2 conditions × 15	P-PD1 vs P-PD2 vs P-PD3 + S-PD1	P-PD1 holds (27/30 rows, verdict fixed): thinking on 12/14 valid, template modal; system line only 0/13 — the reasoning budget is the component

3.1 Registration protocol

For each cell the prompt is pinned and its SHA-256 digest written to disk first — the $N = 13$ square prompt hashes to `32db485b...`. Predictions are registered in the header of each arm's analysis script or in a standalone registration file, with competing predictions, numeric thresholds, tie conventions and a falsifier fixed before the first invocation. Raw outputs are stored verbatim, failures included; scoring is deterministic and local. Which of the square arm's seven cells carry an individually registered point prediction, and which follow from the same closed form without one, is recorded in Appendix J.2.

Four properties are disclosed, not repaired. Temperature, top-p and top-k are not exposed by the agent runtime. The alias-to-weights binding is a promise, not a hash. The subagent inherits instruction files outside the task prompt, so the digests lock a *fragment* of the conditioning context — a replicator cannot reconstruct the inherited files from what we release. Hash coverage is incomplete (§8). Arms P and CL exist because of the first and third of these: they send registered prompts through a serving path that exposes and pins temperature and carries no inherited context (§5.3).

Two scoring conventions were fixed in advance. Validity is reported at 10^{-9} and 10^{-6}, both logged, 10^{-6} primary: proposers print six to eight decimals, so an eight-decimal tangency misses contact by $\sim 5 \times 10^{-9}$, while 10^{-6} sits far below the $\sim 10^{-2}$ gap between rival constructions. Value matching uses a 2×10^{-3} window, which never carries a clearance verdict (§2.4). Every on-prediction rate is computed over *valid* outputs and so conditions on execution success — the variable §4.1 shows can separate what a model attempts from what it lands; per-cell validity denominators are reported beside every rate so the reader can bound the exposure.

Tier	Channel	Instrument	Cells	Clear	Note
weak	direct emission	agent runtime	13, 21, 31, 43	0 of 57	trap cells; at the extend cell $N = 20$ (arm M) 2 of 15 clear, both out-of-family
weak	direct emission	pinned API	7 square cells	0 valid of 105	unscoreable (arm P)
Sonnet	direct emission	agent runtime	13, 21, 31	1 of 30	one out-of-family packing at $N = 31$; 27 of 30 valid at 10^{-9}, the clearance among them (arm S)
weak	math-only program	agent runtime	13, 21, 31	0 of 78	reaches the argmax exactly 2 times, never passes it (CC, CC2)
weak	math-only program	pinned API	13, 21, 31	0 of 12	two cells under the five-valid floor; 0 of 6 at the scoreable cell (arm P code cells)
Sonnet	math-only program	agent runtime	13, 21, 31	0 of 37	annealing and multi-restart search, still 0 (CCS)
Sonnet	math-only program	pinned API	13, 21, 31	0 of 35	the library row's path and 120 s budget without the libraries; best per cell 1.625, 2.1, 2.673, none at the argmax (CCP)
weak	numpy/scipy program	pinned API	13, 21, 31	0 of 11; 0 of 19 pooled	arm CL, then pooled with its registered top-up CL-W (90 invocations; the $N = 21$ cell stays under the five-valid floor): Wilson upper bound 16.8%, under the 20% bar; a post-hoc reparse of the `numpy`-scalar stdouts reads 4 of 41 (CL, CL-W)
Sonnet	numpy/scipy program	pinned API	13, 21, 31	**23 of 25**	3 of 3 cells; excess over the argmax 0.67–4.24%, median 3.62% (CL)

Table 2: **Who clears the family argmax.** Each row is one arm or one registered pair of arms, scored under §2.4's clearance rule: valid at 10^{-9} and above the family argmax by more than 10^{-6}, the argmax recomputed in closed form. "Clear" counts valid outputs. Instrument names the serving path, which arm P shows is not interchangeable: the same weak-tier prompt yields 78% valid packings through the agent runtime and 0 of 105 through the pinned API. Rows are not pooled across instruments.

3.2 The table

Table 2 is the paper. Each row is one arm, or one registered pair, scored under the clearance rule of §2.4; "clear" counts valid outputs whose sum exceeds the family argmax at that N by more than 10^{-6} while valid at 10^{-9}. Four readings.

One cell clears at rate. A Sonnet-tier program allowed `numpy` and `scipy` clears the family in 23 of 25 valid cases, at 3 of 3 cells, its best program per cell by 2.5%, 3.6% and 4.2% of the argmax and its median clearance by 3.6% (§3.5). Against the published best-known sums the paper's bound table carries, the best programs sit 0.5% and 0.9% below at $N = 13$ and 21: above the grid family, not at the frontier.

Neither factor alone, at rate. A weak-tier program with the same libraries clears 0 of 11 in arm CL and, pooled with its registered top-up (arm CL-W), 0 of 19 under the registered parser, a Wilson upper bound of 16.8% that excludes the registered 20% bar, though its $N = 21$ cell holds only 3 valid outputs and stays under arm CL's five-valid floor; under the post-hoc lenient reparse it is 4 of 41, a point estimate of 9.8% whose upper bound of 22.5% does not exclude the bar. The weak tier clears at rate under neither reading, and the exclusion belongs to the registered one alone. A Sonnet-tier program without the libraries — simulated annealing over hand-rolled generators, Halton multi-restarts, force-directed relaxation — clears 0 of 37 (§3.4), and on the library row's own serving path and 120-second budget clears 0 of 35 without reaching the argmax (arm CCP, §3.5). Every weak-tier cell is 0 at the trap cells under the registered scoring.

Escapes exist and are counted. Two cells show low-rate escapes, both out-of-family constructions found by sampling rather than in-family search: one Sonnet direct emission in 30, at $N = 31$, and two weak-tier emissions in 15 at the extend cell $N = 20$ (§3.3). They are in the table because a zero that hides them would be false; they do not change which cell clears at rate.

The instrument column is not decoration. Arm P (§5.3) shows the same weak-tier prompt yields 78% valid packings through the agent runtime and 0 of 105 through the pinned API, so rows are never pooled across the two paths. The clearing cell ran on the pinned path; the Sonnet math-only cell ran on the runtime, and under a 10-second execution budget against the library arm's 120 seconds. Within the weak tier the library comparison is same-instrument (0 of 12 math-only and 0 of 11 with libraries, both pinned, 0 of 6 at the scoreable cell in each; with the top-up 0 of 19 pooled, and a post-hoc reparse reads 4 of 41, §3.5) but still crosses the budget (10 against 120 seconds). Within the Sonnet tier the isolating arm has run: arm CC's math-only prompt on the pinned path under the library arm's 120 seconds clears 0 of 35 and never reaches the argmax (arm CCP, §3.5), so the 0 of 37 to 23 of 25 gap is attributed to the library, on one path and one budget.

3.3 Direct emission

The weak tier emits a computable template. The Haiku-tier proposer produces uniform grids of radius $1/(2k)$ truncated to N circles, with interior-vertex fillers only when the grid underfills. Its modal output is the closed form's prediction at all seven tested N, and at the four discriminating cells — $N = 13$, 21, 31, 43, where prediction and family argmax differ — 41 of 57 valid emissions land on the prediction and 2 reach the rival argmax, both at $N = 21$. No valid emission at those cells clears the family. The per-cell table, the template-shape null it survives, the held-out replication at five N absent from every scoreboard, and the perturbed-container and paraphrase probes are Appendix J.

Two weak-tier emissions do clear the family, and they are the reason the direct row carries a second entry. At the extend cell $N = 20$ (arm M, Appendix J.4) the registered prediction $V(4, 4) = 2.2071068$ was disconfirmed — the model emits the 5×4 rectangle, $T(5, 20) = 2.0$, in 12 of 15 — and two of the fifteen valid rows are staggered rows of four at $r = 1/9$, sum 2.2222222, a hexagonal layout the family does not contain, valid at 10^{-9}. They are not the modal output, they were found by a post-hoc sweep disclosed as such, and they are counted.

The middle tier perturbs and mixes. The Sonnet-tier arm was preregistered in `arm_s_preregistration.txt`, disclosing that 5 of 20 samples at $N = 13/21$ had been seen before registration and that $N = 31$ was fully blind. It was valid 30/30 at 10^{-6} and 27/30 at 10^{-9}, and almost never on-prediction: 0/10, 0/10, 1/10, against the weak tier's 12/16 at the same cells. Instead of truncating a uniform grid it perturbs one, with enlarged edge rows, hexagonal interior rows and two or three distinct radii in the same packing (29/30 multi-radius; Figure 1). One sample at $N = 31$ emitted 27 circles at $r = 1/12$ with 4 corner circles at $r = 1/8$, summing to 2.7499999991, above the family's 2.7485281; recomputed from its coordinates the construction is tangency-tight, with minimum pairwise and wall slack exactly zero, and it is among the 27 samples valid at 10^{-9}. It is the direct-emission row's one clearance: 1 of 30 at the primary tolerance, 1 of 27 at 10^{-9}.

That sample also exposed a scoring defect, and it is why §2.4 states the clearance rule once. At $N = 31$ the family rival (2.7485281) and this out-of-family value (2.75) differ by 1.47×10^{-3}, inside the registered 2×10^{-3} window, so the value classifier counted the escape as a hit on the construction it exceeded. The window was chosen for the anchor comparison, where the nearest competing value is $\sim$0.15 away. Classified by structure instead, Sonnet's rival count at $N = 31$ is 2/10 and pooled 5/30. A third alias, addressed through a bare tier name we call `opus_alias`, was valid in 4 of 30 and is reported in Appendix G with the ladder it completes.

3.4 Math-only programs: an executed program does not escape the family

The code-free design rested on a rationale the first draft asserted and did not measure: that a code channel routes construction to an executed optimizer, so the emission distribution would reflect the optimizer rather than the model. Deployed discovery loops elicit *programs*, so that gap was the standing objection to loop

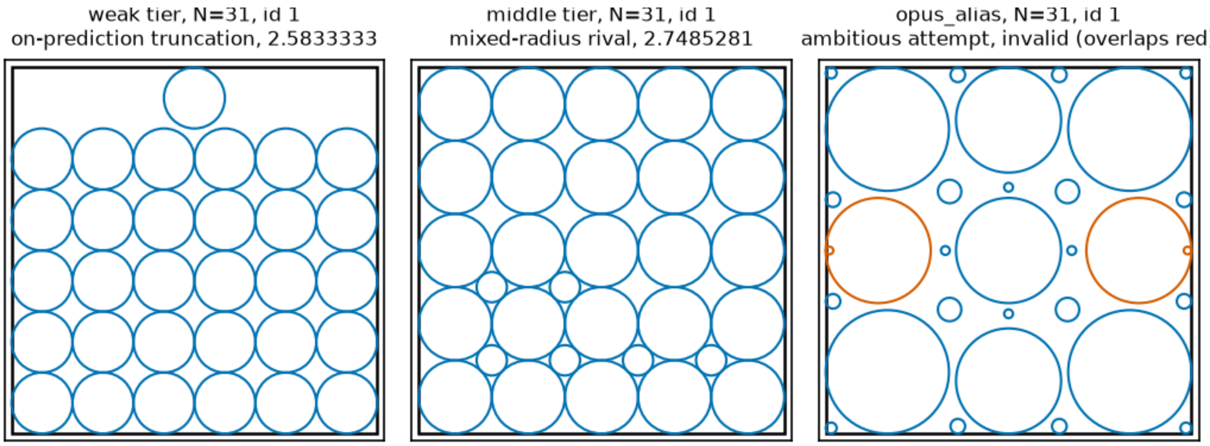

Figure 1: *Three attractor families at a single cell, $N = 31$.* One representative packing per tier, ledger sample id in each panel title (weak tier `bare` id 1, on-prediction truncation 2.5833333; middle tier `sonnet_bare` id 1, mixed-radius rival 2.7485281; `opus_alias` id 1, invalid, with only the offending circles highlighted). Source: `arm_f_candidates_v2.jsonl` via `fig_scripts.py`, deterministic.

relevance. Arm CC measures it. The design was registered and pushed before sampling: the three trap cells $N = 13, 21, 31$, $n = 15$ weak-tier invocations per cell, prompt equal to the bare stem with exactly two registered line replacements — the code-free clause becomes a program contract (standard-library `math` only, stated in the prompt) and the output clause asks for program source. Programs run sandboxed (`python -I`, 10-second timeout, registered AST allowlist; the executor and its four-bin smoke test were committed with the registration), and stdout is scored by arm-F conventions unchanged. Two competing predictions: P-CC1, the modal valid executed output remains the $T(k^*, N)$ bucket at ≥ 2 of 3 cells; P-CC2, it sits strictly above the anchor at ≥ 2 of 3; ties or mixed modes count as neither.

N	Valid	Modal bucket (count, margin)	Anchor-rate	Above anchor
13	12/15	anchor 1.6250000 (2/12, one sample)	2/12	2 — 1.6614054, and the family argmax exactly
21	13/15	anchor 2.1000000 (9/13, margin 8)	9/13	0
31	12/15	anchor 2.5833333 (3/12, one sample)	3/12	0

P-CC1 holds as registered (3 of 3 cells), and its margins belong next to it. At $N = 21$ the mode is heavy — nine exact 2.1000000 emissions, most programs writing the 5×5 grid truncated to 21 circles — but at $N = 13$ and $N = 31$ the modal bucket is the anchor value with a **one-sample margin** over dispersed sub-anchor values, so the modal-identity claim is thin at two of three cells. The robust result is the registered decomposition row: **no valid program output exceeds the family argmax anywhere** (0 of 37 above the rival), and only two exceed the anchor at all, both at $N = 13$ — one at 1.6614054, and one program that computes the family argmax construction $V(3, 4) = 1.7761424$ *exactly.* Failure taxonomy: 7 of 45 programs produce geometrically invalid packings, 1 imports `random` against the stated contract (caught by the registered gate), none time out. What the channel changes relative to direct emission is dispersion, not escape: code-free anchor concentration at these cells runs 56–80%, the code channel's pooled anchor-rate is 14/37, and the lost mass sits entirely *below* the anchor — greedy radius-growth heuristics converging under the lattice.

Both registered follow-ups hold, and they harden the result. A fresh-draw replication wave (arm CC2) and a Sonnet-tier probe (arm CCS) were registered together, after CC's results were known and disclosed as such, 45 invocations each on the byte-identical prompts. **CC2 returned REPLICATED** under its pre-stated verdict map: 0 of 41 valid outputs above the rival (the registered zero-count prediction, exactly), and the anchor bucket modal at 3 of 3 cells — including $N = 31$, where the mode that was one-sample-thin in CC arrived at 10 of 15 valid with a margin of 8. Pooled over both weak-tier waves, 0 of 78 valid program outputs exceed the family argmax and exactly two reach it, one per wave, both at $N = 13$. **CCS returned P-CCS2**: against competing registered predictions (escape at $\geq 20\%$ above-rival versus a zero-count ceiling), **0 of 37 valid Sonnet-tier outputs exceed the argmax**. The Sonnet programs are qualitatively different search — simulated annealing over hand-rolled linear-congruential generators, Halton-sequence multi-restarts, force-directed relaxation, against the weak tier's greedy radius growth — and the tier barely anchors (5 of 37 on the anchor value), yet its executed math-only search never passes the family ceiling at any cell. Taxonomy: CC2 41 valid, 3 geometry-invalid, 1 blocked import; CCS 37 valid, 7 geometry-invalid, 1 timeout. One protocol note, disclosed: a single CC2 invocation ($N = 13$, slot 2) used runtime tools despite the no-tools dispatch wrapper; per the arm-M convention its verbatim final message is scored like every other row. Ledgers and scripts for these arms are listed in Appendix I.

3.5 Library programs: the channel with numpy and scipy

Arm CC's registration named the arm it was not: a library-enabled code channel, the one deployed loops permit. Arm CL is arm CC's registered prompt with a single clause changed — the import allowlist widens from `math` to `math`, `numpy` and `scipy` — run at two tiers (the vendor's `claude-haiku-4.5` and `claude-sonnet-4.5` aliases, temperature 1.0, the same pinned path as arm P) at the three trap cells, $n = 15$ per tier and cell, 90 invocations of which 4 (all Sonnet-tier) were refused by the serving path with an HTTP 402 and are recorded as such, 85 returned a parseable program and one Sonnet response failed to parse, each program executed in a fresh `python -I -S` subprocess with a 120-second wall clock and scored by arm F's conventions with clearance decided only by §2.4's rule. One execution-mechanism correction is disclosed in the registration itself: an earlier draft of it said the libraries would reach the child through `PYTHONPATH`, which `-I` ignores, so a fixed driver committed with the analysis script inserts the interpreter's site-packages into `sys.path` and runs the model's source unmodified via `runpy`; the correction was made before any sampling, in the same commit as the analysis script, and the driver is identical for every row and not part of the scored source. Competing predictions were registered per tier: P-CL1, no valid output clears the family; P-CL2, at least 20% do; and a falsifier F-CL1, the Sonnet tier clears at two or more of the three cells, with the integration rule for that outcome written down before it was known.

Weak tier: the ceiling holds, underpowered. 11 valid programs of 45 (3, 2 and 6 by cell), every one calling `scipy.optimize`, and none clears the family. The 34 failures are 22 programs whose stdout the registered parser could not read, 7 geometrically invalid packings, 4 execution errors and 1 blocked import; none timed out. P-CL1 holds as registered. Under the arm's own five-valid floor only $N = 31$ is scoreable, so the row is 0 of 6 at one cell, with 0 of 11 pooled beside it; the Wilson 95% upper bounds are 39.0% and 25.9%, both above the 20% bar P-CL2 set. The arm can say the weak tier did not clear, not that it cannot at the registered rate; arm CL-W, below, powers the cell.

Post hoc: what the 22 unreadable outputs contain. Every one of the 22 parse failures prints its packing as `numpy` 2.x scalar reprs, `np.float64(0.49...)`, which the registered parser's `literal_eval` rejects; no Sonnet-tier program did this. Stripping that wrapper, and nothing else, parses all 22 (`cl_lenient_reparse.py`, over a re-execution that reproduced every registered bin; 11 of the 22 call `numpy.random` unseeded, so their sums are this execution's): 9 are valid at both tolerances (3, 4 and 2 by cell), 13 are invalid packings (7 overlaps, 4 outside the square, 2 non-positive radii), and **one clears**, a deterministic $N = 13$ program with 13 distinct radii at 1.812531608, 2.0% above the argmax. Under this reading the weak tier is 1 of 20 pooled and 1 of 6 at $N = 13$, a rate inside the registered dead zone (above 0, below 20%). The registered count stands as the arm's result and the table's entry; the lenient count is reported beside it because a zero that depends on a print format would be the wrong thing to lean on.

Arm CL-W: the top-up. Registered after the reparse (`arm_clw_preregistration.txt`, commit `f73222a`, pushed before the first call) with the pooling rule, both readings and the predictions fixed in advance: arm CL's weak-tier prompt byte-identical, same path and decoding, 45 more invocations (15 per cell; the draft said 90 and the credit allowed 45, disclosed), pooled with arm CL's 45 into one 90-row unit scored in one pass by arm CL's pipeline. Of the 45 new rows 8 are valid (3, 1 and 4 by cell); the failures are 23 unreadable stdouts (again `numpy` scalars), 4 invalid packings, 3 execution errors, 3 blocked imports, 3 timeouts and 1 with no program. Pooled, under the registered parser: **0 of 19 valid clear** (6, 3 and 10 by cell, all valid at 10^{-9}; the $N = 21$ cell stays under the five-valid floor and claims nothing on its own), Wilson 95% upper bound

16.8% on the pooled count, so the weak library cell now excludes the registered 20% bar; P-CLW1 holds and F-CLW1 does not fire. Under the lenient reading: 41 valid and **4 clear**, 9.8% (upper bound 22.5%), inside the registered dead zone — at $N = 13$ 1.784872559 and 1.812531608, at $N = 21$ 2.331990574, at $N = 31$ 2.851728045, that last a sum arm CL's Sonnet tier also reached. An independent re-scoring with its own parser reproduced every bin and three of the four sums to the digit; the 1.784872559 program is unseeded and its sum moves with the draw. The weak tier does not clear at rate under either reading.

Sonnet tier: the ceiling falls. 25 valid programs of 45 at the primary 10^{-6} (9, 11 and 5 by cell), every one calling `scipy.optimize`, and **23 clear the family**, each of the 23 valid at 10^{-9} as the clearance rule requires; the two non-clearing programs, both at $N = 13$, are also valid at 10^{-9} (sums 1.5 and 1.767686476, under the argmax), so the count is 23 of 25 at either tolerance: 7 of 9 at $N = 13$, 11 of 11 at $N = 21$, 5 of 5 at $N = 31$, the best program per cell reaching 1.820699211 against the argmax 1.776142375, 2.340549845 against 2.258883476 and 2.864990189 against 2.748528137 — 2.5%, 3.6% and 4.2% above what the recipe family can express. Those are best-of-cell margins; across the 23 clearing programs the excess over the argmax runs from 0.67% to 4.24% with a median of 3.62%. P-CL2 holds and **F-CL1 fires** at 3 of 3 cells. Against the published best-known values the paper's bound table carries (1.829 at $N = 13$, 2.362 at $N = 21$; the table stops at $N = 30$, Appendix J.1), the best programs sit 0.5% and 0.9% below; no bound exists to compare at $N = 31$.

Arm CCP: the isolating arm. Between the 0 of 37 cell (arm CCS, agent runtime, 10-second budget) and the 23 of 25 cell the serving path, the dispatch context and the execution budget changed with the import allowlist, so neither cell alone said which factor cleared. The arm that answers the objection two outside reads made was registered after them (`arm_ccp_preregistration.txt`, commit `ebf213e`, pushed before the first call): arm CC's math-only prompt, byte-identical, on arm CL's path (Sonnet tier, pinned API, temperature 1.0, no system prompt) under arm CL's 120-second budget, with `numpy` and `scipy` blocked by the AST gate. Predictions were competing: P-CCP1, no valid output clears (the library is what clears); P-CCP2, clearance at 20% or more of valid outputs at two of three cells (path or budget explain arm CL), with the falsifier F-CCP1 rewriting contribution 1 to "capability plus budget" if P-CCP2 held. 45 invocations (one in-flight refusal at row 24, resumed with one worker, request unchanged). 35 valid at both tolerances (10, 14 and 11 by cell; 10 geometrically invalid; no timeouts, no blocked imports) and **0 clear**. None reaches the argmax: the best program per cell sums to 1.625, 2.1 and 2.672523443 against 1.776142375, 2.258883476 and 2.748528137, and at $N = 13$ and $N = 21$ that best is the template anchor $T(k^*, N)$ itself, with the anchor's structure modal (7 of 10 programs at $k = 4$, 12 of 14 at $k = 5$). P-CCP1 holds; F-CCP1 does not fire. The three Sonnet program cells read 0 of 37 (runtime, 10 seconds), 0 of 35 (pinned, 120 seconds) and 23 of 25 (pinned, 120 seconds, libraries): within the Sonnet tier, path and budget moved nothing and the libraries moved everything. At the weak tier the same libraries move nothing at rate (arm CL-W, above), which is why §1 says leaving the family at rate took the tier *and* the optimizer.

What the arms establish. The family ceiling holds at rate in the math-only channel and at the weak tier. It survives conditioning, held-out cells and executed arithmetic without libraries, at both tiers, on both serving paths and at both budgets; it does not survive a library-enabled program written by a middle-tier model. With arm CCP the attribution is no longer crossed within the Sonnet tier: the library is the factor. The 10-second budget of arms CC, CC2 and CCS remains a difference from the library arm for the weak tier's own contrast, and is disclosed as such in §7. Per the registered integration rules, the abstract and title name the library-enabled Sonnet tier as the regime that clears, arm CCP's 35 valid programs are reported here by name and not pooled with the 0 of 37, and the pooled ceiling of §3.6 keeps the library arm outside the pool by named regime. Ledgers and scripts are listed in Appendix I.

3.6 The ceiling as a row: 0 of 290

The table's zeros pool with three arms outside it that test the same value in regimes the direct rows do not cover — held-out cells, one conditioning step, five loop generations — and the pooled statement is the one an evaluation reader can use:

corpus	regime	clears	where
115 valid program outputs (CC 37, CC2 41, CCS 37)	executed math-only code, two tiers, fresh-draw replication	0	§3.4
63 valid outputs	held-out N, five cells absent from every scoreboard	0	Appendix J.6
49 valid conditioned outputs	five generations of a registered archive-and-mutate loop	0	§5.2
63 valid conditioned outputs	one conditioning step, three parent conditions (10^{-9})	0	§5.1
290 valid outputs (281 at 10^{-9} throughout)	CC, CC2, CCS, CN, MU, L	**0**	—
12 valid program outputs	math-only code, bare pinned API path (not pooled)	0	§5.3
35 valid program outputs	math-only code, Sonnet tier, pinned path, 120 s (not pooled)	0	§3.5
19 valid program outputs	numpy/scipy code, weak tier, arms CL and CL-W pooled (not pooled here)	0	§3.5
25 valid program outputs	numpy/scipy code, Sonnet tier (not pooled)	**23**	§3.5

Pooled, that is **0 of 290 valid outputs** across six arms, selected by regime and not by outcome: math-only programs at two tiers, cells no scoreboard lists, a conditioning step, an iterated loop. Three things travel with the number. It is not one tolerance: the first three rows count outputs valid at 10^{-6}, the arm MU row counts at 10^{-9}, and arm MU is the one arm where the legs disagree — held at 10^{-6} throughout, the pool reads 315 valid outputs with 18 above the argmax, all 18 from arm MU and every one failing 10^{-9} because the excess is its own overlap (§5.1). Held at 10^{-9} throughout, which is the tolerance the clearance rule requires, the pool reads 281 valid outputs with 0 above: the three code arms are unchanged (37, 41 and 37 valid at both tolerances, re-scored from the stored programs through the registered scorer), CN 56 of its 63, L 47 of its 49, MU 63. The recount is post hoc bookkeeping over rows already in the study (`recount_1e9.py`); it adds no invocations and moves no verdict. It is not all registered: arms CC/CC2/CCS registered zero-count predictions and arm L's L4 was registered open, but the clearance question was put to arm MU's rows post hoc; arm CH's 14 valid outputs, conditioned on a score table rather than on a parent, are held out of the parent-conditioning row and none of them clears either. And it is not a bound: the three direct-emission escapes of §3.3 and the twenty-three library programs of §3.5 sit outside it by named regime, and the middle-tier escape sits at $N = 31$, a cell five of the six pooled arms also sample. What the zero says is that on this benchmark, at these cells, a weak-tier proposal or a math-only program at either tier does not exceed a value computable in advance, so a discovery claim from those proposers has a number to beat before any search is credited.

4 Mechanism: choice follows the score table, execution does not

Table 2 says a weak-tier proposer leaves the family at rate in no channel and a Sonnet-tier one leaves it at rate only in the library cell. Two readings of that are open: the weak tier *prefers* the template, or it *cannot build* anything else. One registered arm separates them, and three other arms show the same split from different directions.

4.1 Arm CH: the argmax in hand does not dislodge the template

Arm CH hands the model the entire recipe family: the bare prompt plus every admissible $T(k, N)$ and $V(k, m)$ value enumerated, including the family argmax, with the instruction to use whichever scores highest or anything better ($n = 15$ per cell at $N = 13$, 21, 31; 45 invocations; registered together with arm MU before sampling, prompt hashes in `arm_mu_prompts.json`). The §7 tractability alternative says the model truncates at k^* because that is the only construction it can compute; enumerating the family removes the computation burden, so under that reading the modal output should become the stated argmax (P-CH2). Under the template reading it stays $T(k^*, N)$ despite the argmax being printed in the prompt (P-CH1).

Registered verdict: P-CH1 holds (2 of 3 cells) — but the registered metric carries a survivorship artifact we disclose rather than exploit. Under the registered scorer, which takes the modal output *among valid samples*, the template reading wins: at $N = 13$ the modal valid output is $T(4, 13) = 1.6250000$

against a stated argmax of 1.7761424, at $N = 31$ it is $T(6, 31) \approx 2.5833$ (4 of 7 valid) against 2.7485281 (3 of 7) — a one-sample margin — and only $N = 21$ goes to the argmax (2.2588835, 3 of 5 valid). P-CH2 holds at only 1 of 3 cells. This arm sets no evaluability floor of its own, so it grants $N = 13$ a per-cell verdict on two valid samples; under the GM chain's floor of three that cell would not be scoreable and P-CH1 would hold at 1 of 2. The floors differ because each was fixed at its own arm's registration, and we report both readings rather than harmonize them afterwards (§2.4). Read no further and the tractability alternative fails the probe. But validity is low in this arm — 2, 5 and 7 valid of 15 by cell — and the failure modes change the picture at the *attempt* level.

The attempt-level reading, post hoc and disclosed. Of the 31 invalid rows, 30 are evaluable, and **all 30 attempt the argmax at its own grid order** k^*-1: 12 of 13 at $N = 13$, 10 of 10 at $N = 21$, 8 of 8 at $N = 31$, with the order backed out of each row's dominant emitted radius the way Appendix J.2 recovers k. The thirty-first row emits radical literals the scorer cannot evaluate and is therefore not scored as attempting anything. Attempting is not the same as carrying a family filler radius, which would not identify which k was tried; the order does. Of the 30, all misplace the fillers — typically at the container corners, where radius $(\sqrt{2} - 1)/(2k)$ overlaps the adjacent grid circle, against the correct construction's grid interstices.

The control this reading needs. Attempt-level counts mean nothing without knowing what the model attempts when *nothing* hands it the argmax. Arm F's bare invocations at the same three cells are that condition, and the same dominant-radius estimator applies to them. Over *all* evaluable rows, valid and invalid alike, the bare arm reaches for the argmax order in **5 of 57** (9%) against arm CH's **36 of 44** (82%); Fisher's exact test gives $p = 2.2 \times 10^{-14}$ one-sided (`arm_f_attempt_uncond.py`, post hoc, disclosed). We report that comparison rather than the invalid-row one because the invalid-row version conditions on a variable the manipulation itself moves: arm CH is 31 of 45 invalid, arm F 10 of 60 at the same cells, so restricting both arms to their invalid rows conditions on a collider. For the record it points the same way — 3 of 7 evaluable invalid bare rows at the rival order against arm CH's 30 of 30, $p = 5.3 \times 10^{-4}$ (`arm_f_attempt_control.py`) — but it is a diagnostic, not the evidence. Both comparisons are post-hoc strata rather than registered outcomes.

What the arm supports. Counting attempts rather than valid outputs, the model reaches for the argmax in 36 of 45 CH invocations — the 30 evaluable invalid attempts plus 6 valid outputs that land it; the control above quotes 36 of 44 because the forty-fifth row, the radical-literal one, is not evaluable and is excluded from both arms' denominators there. It also *builds* it: 0 of 2 valid at $N = 13$, 3 of 5 at $N = 21$, 3 of 7 at $N = 31$, 6 of 14 pooled. So the model does not *fail* to build the argmax; it builds it unreliably, in 6 of 14 valid outputs, while reaching for it in 36 of 45 invocations. The registered metric was fixed before sampling and we report its verdict as registered — and flag in the same breath that it conditions on exactly the variable, execution success, that separates the two readings. The honest summary is a decomposition rather than a clean winner: handed the argmax, the model's *choice* follows the score table, and its *execution* succeeds less than half the time it is attempted — the tractability alternative is wrong about selection and right about instantiation.

N	Stated argmax	Modal valid output	Valid	Invalid rows attempting argmax	Registered verdict
13	1.7761424	$T(4, 13) = 1.6250000$	2/15	all	P-CH1 holds
21	2.2588835	argmax, 3/5	5/15	all	P-CH2 (argmax)
31	2.7485281	$T(6, 31) \approx 2.5833$ (4/7)	7/15	all	P-CH1 holds (one-sample margin)
pooled	—	template 2 of 3 cells	14/45	**30/30** evaluable (all misplace fillers; 1 further row unevaluable, radicals)	choice follows score table (36/45 attempt it); execution lands it 6/14 valid

4.2 The same asymmetry from three other directions

In code. Delegating the arithmetic to an executed math-only program is the cleanest removal of the hand-computation burden, and it recovers the argmax construction exactly twice in 78 valid weak-tier programs, one per wave, both at $N = 13$ — a program that computes $V(3, 4) = 1.7761424$ — and never passes it (§3.4). The lost probability mass sits below the anchor, not above it: greedy radius-growth heuristics converging under the lattice. With `scipy.optimize` available the weak tier still clears nothing in 19 valid programs under the registered parser (4 of 41 under a post-hoc lenient one), while the Sonnet tier clears in 23 of 25 and, without the libraries on the same path and budget, in 0 of 35 (§3.5): the library supplies execution the weak tier does not wield.

At a held-out cell. Arm CN samples the bare template at five N absent from every cited scoreboard (Appendix J.6). Its one miss, $N = 50$, is the arm-CH pattern in the wild: 10 of 11 valid outputs are the registered 7×7 grid plus exactly one filler, 4 with the filler at the interstitial radius $(\sqrt{2} - 1)/14$ (the registered value) and 6 with it pushed into a container corner at $r \approx 0.0123$ — the cell's mode, 0.017 below the prediction. The template is selected; the filler is mis-executed; the value criterion registers a miss, and it is reported as one.

Across a vendor. A second weak-tier family, gemma-4-26b under a laboratory-grade instrument, does what the original does not: at every discriminating cell its modal output is the rival argmax itself, and a post-hoc structural diagnostic finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, 27 of 43 discriminating-cell validities against the original family's 2 of 57 (§5.4, Appendix J.5). Read against the tractability alternative, the anchor value marks where construction-by-mental-arithmetic lands, and a family with more arithmetic reach executes the harder in-family construction rather than truncating — same recipe family, other branch.

Up the ladder. Holding prompt, container and scoring fixed across three nominal tiers at the three discriminating cells, ambition rises monotonically — truncated template, perturbed hybrid, recursive gasket — while validity rises then collapses, 83% → 100% → 13% on the matched cells (the ladder table and the top alias are Appendix G; the alias-to-weights binding is not attestable). "Ambition" is measurable: a post-hoc diagnostic (`diagnostics_ambition.py`, all samples with emitted geometry, invalid included) gives median distinct radii per sample $1 \to 2 \to 3$ and median off-lattice center fraction $0.08 \to 0.43 \to 0.95$ across the tiers. The top alias attempts at every cell a more ambitious construction than either other tier — quarter-circle corners with Apollonius-style fillers, coarse grids with border strips — and fails geometrically rather than numerically (24 overlaps and 2 nonpositive radii in its 26 failures; Appendix G). The tier that attempts the most builds the least.

What the four directions share is the order of failure. Selection is not the bottleneck: the weak tier picks the argmax when shown it, a family with more reach emits it unprompted, and the ambitious tier reaches for constructions the family does not contain. Execution is: the picked construction is mis-built in text, reached and not passed in math-only code, and cleared at rate only where a mid-tier model hands the arithmetic to an optimizer. The template is what survives execution, and the library-enabled Sonnet cell is where what survives changes.

5 Zero-shot characterization does not transfer

The weak tier's template is a closed form, and Appendix J reports the prediction in full: modal at all seven tested N, at four of five held-out N, under a perturbed container and under paraphrase. A characterization that sharp invites the inference that it describes the proposer inside a loop. Three registered arms each changed one thing about the setting — a parent, a generation, the serving path — and each carried registered predictions for the inference. The persistence prediction failed in each: arm MU's falsifier fired, arm L failed two of its four predictions (a third was registered open), and arm P's predictions were unscoreable because validity collapsed. This section reports them as results, not as caveats, because the same inference is routinely drawn from zero-shot characterizations of LLM proposers.

5.1 Arm MU: the anchor dissolves after one conditioning step

Arm MU issues one-parent mutation calls: the bare prompt plus an existing packing and the instruction to modify it so the sum of radii increases. Three parents per cell ($N = 13$, 21, 31; $n = 15$ per condition-cell, 135 invocations): *A_anchor*, the model's own unconditioned anchor $T(k^*, N)$; *B_rival*, the provably better in-family rival $V(k^*{-}1, m)$; and *C_offfamily*, a diagonal-chain packing scoring ≈ 0.59–0.62, far below either. It was registered together with arm CH before sampling, with numeric decision thresholds, a power analysis

(0.83 at the registered effect size), tie-against conventions, and a falsifier (F-MU1) whose consequence — the anchoring claim is scoped to *unconditioned* calls only — was written down before the first invocation.

F-MU1 triggered; the anchoring result dissolves under conditioning. Under B_rival the pooled anchor-rate is 0 of 26 valid (0%, CI $[0.00, 0.13]$) against a registered DISSOLVES threshold of $\leq 20\%$; keep-or-improve is 26 of 26 against a threshold of $\geq 50\%$. Every valid B_rival sample keeps or refines the rival construction, and every one inherits the rival's grid order (k-inheritance 26/26, disconfirming P-MU3's $< 50\%$ prediction). The template does not reassert itself when a better in-family parent is present. A_anchor makes the same point from the other side: even with the model's *own* anchor as parent and an explicit increase-the-sum instruction, only 7 of 28 valid samples (25%, CI $[0.13, 0.43]$) stay on the anchor — P-MU1's $\geq 50\%$ prediction is not met. The one condition where the template does reassert itself is the one the registration predicted it would: given the *off-family* diagonal parent, 20 of 34 valid samples (58.8%, CI $[0.42, 0.74]$) abandon the parent entirely and emit a k^*-grid construction — P-MU4 holds per-cell at $N = 21$ (7/13) and $N = 31$ (12/14), though not at $N = 13$ (1/7). A good in-family parent is followed, a bad off-family parent is discarded for the template: the attractor is real but weak, capturing the trajectory only when the parent is worse than what the template supplies.

Parent	Valid	Anchor-rate (CI)	Registered threshold	Keep-or-improve	k-inher.	Reading
B_rival (in-family argmax)	26	0/26 (0%, $[0.00, 0.13]$)	DISSOLVES if $\leq 20\%$	26/26 (bar $\geq 50\%$)	26/26	**F-MU1 triggered** — anchoring does not transfer to one-parent calls
A_anchor (model's own anchor)	28	7/28 (25%, $[0.13, 0.43]$)	P-MU1 predicted $\geq 50\%$	—	—	not met — even the anchor-as-parent does not hold the anchor
C_offfamily (diagonal, ≈0.6)	34	20/34 abandon parent for k^*-grid (58.8%, $[0.42, 0.74]$)	P-MU4: template reasserts	—	—	holds at $N = 21, 31$; not at $N = 13$ (1/7)

The ceiling holds inside the conditioned regime (post hoc, disclosed). Arm MU's rows were scored against the *anchor*, so whether any conditioned output clears what the family can express was never asked of them at registration. It is asked here. Recomputing the family argmax from the closed form — 1.776142375 at $N = 13$, 2.258883476 at $N = 21$, 2.748528137 at $N = 31$ — arm MU's 135 invocations give 88 valid outputs at the primary tolerance and 63 at 10^{-9}. Eighteen of the 88 sit above the family argmax by more than the 5×10^{-8} granularity at which their own sums are recorded, by at most 1.5×10^{-6}, and *every one of the eighteen fails the* 10^{-9} *test*: they are packings whose overlap is small enough to pass the loose gate, and the excess is the overlap. Twelve further rows exceed the argmax only within that recording granularity, which is to say they are the argmax, written down. Arm CH's 45 invocations add 14 valid outputs at both tolerances, none above. The verdict rests on the 10^{-9} leg, where none clears, and it is corroborated at structure level: each of the eighteen carries an overlap the tighter gate rejects, and the excess is bounded above by the overlap. This re-reading adds no invocations to the study total; the recomputation is `arm_mu_ceiling.py`.

5.2 Arm L: iterating the loop moves the anchor but not the ceiling

Every arm to this point samples once. The regime that discovery loops actually run in iterates: a proposal is scored, kept or discarded, and fed back as a parent. Arm L runs the minimal generational loop and reports the four registered predictions whether or not they held. The design is registered before sampling, with both prompt templates hashed and the stepper and scorer committed alongside. Generation 0 is the bare template A.1, byte-identical to arms F, M and CN; generations 1–5 are arm MU's registered mutation template verbatim, instantiated with the parent packing and its score. Two discriminating cells, $N = 13$ and $N = 31$, are crossed with two archive regimes — GREEDY, which retains only the single best valid proposal, and DIVERSE, which retains up to five distinct-valued ones — giving four lineages of six rounds at population 5: 120 invocations, of which 2 were runtime-rejected and counted. The parent for population slot i is archive[i mod |archive|], so the loop carries no sampling randomness of its own. All four lineages are evaluable under the registered floor. This is a minimal analogue and the registration says so: population 5, six rounds, one-parent mutation, no islands, no crossover, no program representation, and no evaluator signal beyond the parent's own score. Nothing here transfers to deployed systems, and no claim about any named system follows.

L2 confirmed; L1 and L3 not met. L2 (dissolution survives iteration) holds in both cells: the on-prediction rate falls from 4 of 9 valid generation-0 proposals to 5 of 20 conditioned ones at $N = 13$, and from 7 of 8 to 6 of 29 at $N = 31$. The falsifier F-L1 did not fire. L1 (the registered prediction is modal at generation 0 in every lineage) is **not met**, on one lineage: the 13-DIVERSE seed round returned four valid proposals split two on the prediction and two elsewhere, and the registered tie rule counts the tie against it. The other three lineages pass. The failure is a power artifact of the registration — we set four separate per-lineage bars over roughly four valid proposals each rather than one pooled bar per cell — and it is reported as scored, not rescued. L3 (loop diversity is scaffold-produced, so the greedy archive should return proposals toward the family) is **not met, with the direction reversed**: the conditioned on-prediction rate is 3 of 22 under GREEDY against 8 of 27 under DIVERSE. The comparison is confounded — greedy and diverse lineages differ in parent identity as well as archive width, and 13-GREEDY produced only 4 valid proposals across five conditioned rounds — so the reversal is reported as a disconfirmation of the registered prediction and not as evidence for its converse.

L4: the ceiling holds under iteration. L4 was registered open, with no direction predicted: report best-of-run against the family argmax. **No lineage clears it.** At $N = 13$ both lineages top out exactly at the family argmax 1.7761424; at $N = 31$ GREEDY reaches 2.6104167 and DIVERSE lands exactly on the family argmax 2.7485281. Across 49 valid conditioned outputs, zero exceed the best value the recipe family can express. Iteration moves the proposer off the anchor without moving it past the family *in these arms*. One correction is disclosed: the registered configuration stored the family argmax to seven decimals, and the clearance test compares at 10^{-9}; a proposal landing exactly on the rival at $N = 31$ therefore tested as *above* the truncated literal by 3.7×10^{-8}. The scorer now recomputes the argmax in closed form, agreeing with the stored value to 1.3×10^{-15}; the change is post-sampling and affects only the 10^{-9} clearance test (corrections ledger item 34). Scale is the honest limit: five conditioned generations, population 5, 49 valid conditioned outputs, one tier. The arm shows that this loop, at this size, on this channel, does not clear the family; it cannot show that a larger one would not. Raw rows in `arm_l_collect.jsonl`; frozen report `arm_l_report.json`, scored once on the complete ledger. The study corpus stood at 1068 invocations at that point.

5.3 Arm P: the registered prompt on a bare, pinned serving path

Every direct-emission arm above ran through an agent runtime whose decoding parameters are not exposed and whose conditioning context the prompt digests lock only in part (§3.1). An external referee named that the paper's largest instrument hole. Arm P answers it the only way it can be answered: the registered prompts, byte-identical — the builder asserts every hash on file, and gives $N = 43$ its first — sent through a serving path that exposes and pins temperature (OpenRouter chat-completions, the vendor's own `claude-haiku-4.5` alias, temperature 1.0, no system prompt), at $n = 15$ per cell across the seven square cells, the five held-out cells and the three code cells: 225 invocations, 219 returning content and 6 refused by the serving path and recorded as such. It was registered with competing predictions on mode identity (P-P1, P-P2), on the code-channel ceiling (P-P3, P-P4) and a falsifier on the discriminating-cell rate (F-P1), and it is a new arm on a new instrument, reported beside arms F, CN and CC and merged into none of them.

Validity collapsed. At the seven square cells, 0 of 105 outputs are valid at either tolerance. Every failure is an overlap: the responses are well-formed lists of the requested count with no prose and no fences, emitted with no reasoning tokens, and their median maximum pairwise overlap at $N = 13$ is 0.10, fifty times the value window. At the held-out cells, 4 of 75 are valid, all four at $N = 50$ and all four summing to 2.5, a full unit below the family argmax there (3.5295867); none clears. The direct-emission predictions are therefore UNSCOREABLE at every cell under the arm's own floor of five valid outputs, and F-P1's denominator is empty — the registration did not anticipate a collapse, and we report that rather than read a zero either way. On the code cells, arm CC's prompt byte-identical and executed and scored exactly as arm CC's (10-second timeout), the picture is the paper's: 12 valid programs of 45 (4, 6 and 2 by cell, only $N = 21$ above the floor) and none clears the family under §2.4's rule; P-P3 holds.

Two post-hoc diagnostics, disclosed as such. Temperature is not the cause: the $N = 13$ prompt at temperature 0.0, 0.5 and 1.0 gives 0 of 6 valid at each, all overlaps. The instrument is: the same prompt with the registered dispatch wrapper, through the agent runtime's haiku alias that arms F, CN and CC used, six subagents in parallel on the same day, gives 6 of 6 valid at both tolerances and lands the registered anchor $T(4, 13) = 1.625$ in 4 of 6, with one 3×3 grid carrying corner fillers at 1.6144 and one centre-plus-border construction at 1.7500, above the anchor and below the family argmax.

Arm P-D: which component of the instrument. Registered after an outside review asked for a diagnostic beyond temperature (`arm_pd_preregistration.txt`, commit `5554cc0`, pushed before the first call): the same $N = 13$ prompt, same alias, same pinned path and decoding, under two one-line manipulations, $n = 15$ each. D1 turns the vendor's extended thinking on (the routing layer's default effort; median 3160 reasoning tokens per row) with no system prompt; D2 leaves thinking off and adds one system line, "You are a careful assistant. Think before you answer." Competing predictions: P-PD1, D1 valid in at least 8 of 15 and D2 in at most 3 (the reasoning budget is the cause); P-PD2, the mirror image (the context is); P-PD3, anything else. Interim, at 27 of 30 rows (14 D1, 13 D2; the last three await credit and cannot change the verdict, since D2 can reach at most 2 valid and D1 already holds 12): **D1 is valid in 12 of 14** (11 at 10^{-9}; the two failures overlap), **D2 in 0 of 13**, all overlaps, exactly arm P's collapse. P-PD1 holds. And the anchor comes back with the budget: the modal valid D1 output is $T(4, 13) = 1.625$ (5 of 12, the 4×4 grid truncated), $k = 4$ is the modal structure (6 of 12), and one output is the family argmax 1.7761424 itself, the rival the runtime rows reach at 2 of 57. S-PD1 is met.

What the arm establishes. The anchoring result is a property of the instrument that produced it, and arm P-D names the component: the reasoning budget. The same prompt, sent bare to the same vendor's alias with temperature pinned, does not yield valid packings from this model class at all; give the pinned call the vendor's extended thinking and it yields valid packings at the runtime's rate with the template modal, while a system line that asks for care changes nothing. The runtime's undisclosed conditioning context — the fragment §3.1 disclosed the digests do not lock — is not what carries the template; the thinking budget is. That is why Table 2 carries an instrument column and never pools across it, and it is a reproducible instrument, since the budget is a request parameter on the pinned path. Ledgers and scripts are listed in Appendix I.

5.4 Across vendors: a boundary that runs both ways

Two registered arms take the same byte-identical prompts across vendors — arm GM3 under a laboratory-grade instrument against one second-vendor family, arm V under free-tier serving against thirteen aliases spanning seven attributed vendors and one unattributed alias. One table carries the chain's verdicts; GM3's per-cell table and figure are Appendix J.5, and arm V is Appendix F in full.

Vendor	Model	Instrument	Registered test	Verdict
Google	gemma-4-26b	first-party API, 7 $N \times 20$ (41/140 rows routed under amendment; dual accounting)	pooled $\geq 30\%$; cell-level ≥ 5; falsifier ≥ 4 fails	pooled bar met (57.5%); discriminating cells 11/43 = 26%, **below** the bar; cell-level short (modal in 2 of the 5 scoreable cells, against a bar of 5; falsifier not triggered); all three misses are the *rival construction* (27/43)
Cohere	north-mini-code	free-tier alias, 5 $N \times 5$	V1 $\geq 50\%$ of valid on-prediction; V2 rival $\leq 10\%$	**TRANSFERS**, point-estimate (8/12 = 67%; CI includes bar); V2 HOLDS (0/4)
OpenAI (open-weight)	gpt-oss-20b	free-tier alias, 5 $N \times 5$	same	**TRANSFERS**, point-estimate (11/18 = 61%; CI includes bar); V2 HOLDS (0/9)
Google	gemma-4-31b	free-tier alias, 5 $N \times 5$	same	**DOES-NOT-TRANSFER** (0/15); V2 HOLDS (0/3)
10 further aliases	(NVIDIA, Liquid, Poolside, inclusionAI, others; one unattributed)	free-tier	floor: ≥ 10 of 25 invocations valid	BELOW-FLOOR; failure taxonomies logged, no claim in either direction

Arm GM3 re-runs the unconditioned-call protocol against a second vendor's open-weight family (gemma-4-26b-a4b-it) through its first-party API: seven N cells, twenty samples each, registered predictions and falsifier fixed before sampling. The pooled prediction held — 46 of 80 valid packings (57.5%) land within 2×10^{-3} of the value predicted by §2.3's rule, against a registered 30% bar, for a rule fitted entirely on another vendor's models — and the pass rides on the non-discriminating cells: 35 of the 46 sit at $N = 17$ and $N = 35$, where prediction equals the family argmax, and on the three discriminating cells the rate is 11 of 43 (25.6%), below the bar. The cell-level prediction did not hold (modal in 2 of 5 scoreable cells, short of the registered 5) and the falsifier (failure in ≥ 4) did not trigger either; the registration left a dead zone and the outcome landed in it. The three misses are the finding §4.2 uses: the family's modal output at every discriminating cell is the rival argmax itself, 27 of 43 discriminating-cell validities carrying its exact two-radius decomposition. Validity collapsed at the two largest cells (2 then 0 valid of 20), both UNSCOREABLE under the registered rule.

Arm V varies vendor and serving class at once, under the same byte-identical arm-F prompts, across thirteen free-tier aliases at five cells and five samples each (877 attempt rows over 325 registered slots). Ten of thirteen aliases finished below the registered validity floor and claim nothing in either direction. Of the three that cleared it, two transfer at point estimate — 8/12 and 11/18 on-prediction, Wilson intervals spanning the 50% bar — and gemma-4-31b does not, at 0/15. At the cells that can discriminate anchoring from family search the two passes are 3 of 4 and 5 of 9, and neither separates from the template-shape null of Appendix J.2 (exact upper tails 0.111 and 0.145). Zero rival-argmax emissions in 16 discriminating-cell validities.

So the anchor is neither a single-vendor artifact nor universal: it transfers at the pooled level to two further vendors, one weak-tier family takes the rival instead, and a sibling of that family at the same floor anchors on nothing. With arms MU, L and P this closes the boundary from every side the study reached: the template is a property of one tier, one channel, one instrument, and one conditioning regime, and it is inside those four conditions that the closed form names the modal output in advance.

6 Related work

LLM-driven discovery and the circle-packing scoreboard. Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt (Guo et al., 2024), EoH (Liu et al., 2024) and LLaMEA (van Stein & Bäck, 2025) established the LLM call as an EC operator over prompts and programs; FunSearch turned the pattern into a discovery claim, and every system in §1's scoreboard inherits it. Two orderings in that scoreboard matter: HELIX's figure is marginally *below* ShinkaEvolve's relaxed-tolerance figure (and above its strict one) while being described as state of the art, and ThetaEvolve claims new best-known bounds, so the benchmark is saturated only across the AlphaEvolve–ShinkaEvolve–HELIX cluster and the number is still moving. OpenEvolve (Sharma, 2025), the open-source AlphaEvolve reproduction most practitioners run, reports values in the 2.634–2.636 band across its published example runs (2.634292 in the circle-packing example at the time of access; the repository is unpinned). Appendix J.1 records the correction to our record comparison and its $N > 30$ coverage gap. Two critiques bear on our framing and are frequently conflated, so we attribute them explicitly: Gideoni et al. (2026) show simple baselines recover much of the reported advantage, and Berthold et al. (2026) show classical solvers do too.

Template convergence and diversity collapse. "Mutation Without Variation" (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates — in 87% of chains, over 93% of mutations revisit a previously seen form — which is the same bias family in a different regime: mutation loops, program space, no closed form, no preregistration. It is also the closest published neighbour to our arm L, whose registered L2 confirmed the analogous statement inside a scored loop: iterating the archive moves the anchor without lifting the ceiling above it (§5.2). Their result is about structural revisiting; ours is about a value bound that the revisited structures cannot cross. On anchoring, Huang et al. (2026) treats anchoring as a general bias over initial information while Lou & Sun (2024) and Malberg et al. (2025) document numeric primes dragging point estimates; our modality is the difference, since a construction template has a computable value where a scalar anchor has only a measurable pull. Concurrently, "Measuring the Gap Between Human and LLM Research Ideas" (Chen et al., 2026) reports the same collapse one level up, in idea space: across 11,683 paired proposals, LLM research ideas concentrate on bridge-and-synthesis framings (47–64% versus 12% for humans; normalized entropy down to 0.55 versus >0.92), and chain-of-thought pushes the distribution *further* from the human one. That result is distributional; ours is the limiting case where the collapsed distribution's mode is a single closed-form-predictable object — narrowness measurable there, computable here. "Artificial Hivemind" (Jiang et al., 2025) documents the same homogeneity across open-ended domains at survey scale, which makes cross-domain generality of the collapse the expected default; against that backdrop our contribution is not that collapse occurs but that, in this regime, its mode is a single closed-form-computable object, stated before sampling. Converging skepticism about what the proposer contributes comes from "Dictionaries, Not Darwin" (Li, 2026), Wang et al. (2026), BehaveSim (Zhang & Lu, 2026), Strategy Diversity (Yang et al., 2026), the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025) and MathConstruct (Balunović et al., 2025). Two results cut the other way and belong on the other side of the ledger: Zhang et al. (2024) provides empirical grounding for the *importance* of evolutionary search, which is evidence against our substitution of an unconditioned call for the loop and is engaged in §7; and Zhang et al. (2026b) finds strong LLM optimizers act as *local refiners*, which is what a parent-conditioned call would be, so it bears on §7's mutation arm.

Scaling inversions. Zhou et al. (2024) show larger and more instructable models becoming less reliable; the o3/o4-mini system card (OpenAI, 2025) shows the same on PersonQA; Shojaee et al. (2025) reports collapse past a complexity threshold; GeoBuildBench (Kim et al., 2026) finds geometric construction a regime where nominal capability and executed correctness diverge. Our instantiation is *attractor family* versus executability.

Trace faithfulness, and what our result is not. One cluster intervenes *on trace content* and finds answers do not move — Reasoning Theater (Boppana et al., 2026), Project Ariadne (Khanzadeh, 2026), "Beyond the Commitment Boundary" (Scalena et al., 2026) and "The Chain Holds, the Answer Folds" (Li et al., 2026) all report causal decoupling. A second cluster *estimates* faithfulness without observing the process, represented here by Arcuschin et al. (2026); Appendix E.5 reports the case where ground truth for the described artifact exists. Our result sits at a third intervention point: we vary whether a trace is *requested* rather than editing its content, which is measurement reactivity, not causal faithfulness. The closest cousin is "The Price of Format" (Yun et al., 2025), showing format constraints collapse generation diversity — and as Appendix E.1 concedes, our manipulation includes a small output-format change, so we commit a miniature version of the confound we cite. Format-restriction performance costs are documented at scale in Tam et al. (2024). Related observer-effect and elicitation results: (Abdelnabi & Salem, 2025; Zhang et al., 2026a; Wu et al., 2025).

Preregistration lineage. Thomas et al. (2026), Williams et al. (2026), Vaccaro (2026) and Zhang (2026) preregister recipes, outcome-blinded predictions and agent protocols; HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. We adopt the standard rather than claim it. What we add is the object locked — exact closed-form point predictions of specific output values — and, from Appendix E.3, the observation that locked prose and executing scorer can disagree at the boundary. Structural precedents: Kaplan et al. (2020) and Hoffmann et al. (2022) publish a functional form then run the confirming instance; Snell et al. (2024) is the closest structural twin. QDAIF (Bradley et al., 2023) is the QD-descriptor setting most affected by our Appendix E result, and GigaEvo (Khrulkov et al., 2025) the concrete in-scope system running MAP-Elites with LLM-driven mutation, where a self-reported behavioral descriptor would be perturbed in practice.

7 Limitations and untested alternatives

The measured distribution is not the in-loop distribution. The direct-emission rows sample generation-0, parent-free, feedback-free calls; the systems in §6 sample parent-conditioned program proposals under selection, and Zhang et al. (2024) is published evidence that evolutionary search contributes materially. Arm MU (§5.1) measured the first conditioning step and its falsifier fired — 0 of 26 valid rival-parent samples return to the anchor — so the anchoring result is scoped to unconditioned calls and the table's direct rows carry that scope.

The limitation has since narrowed rather than disappeared. Arm MU samples one conditioning step; arm L (§5.2) samples five generations of a registered archive-and-mutate loop, which is the in-loop distribution this limitation names. Both were then read against the family ceiling rather than only against the anchor, and the tolerance has to be named because the two legs disagree in arm MU: at 10^{-9} no conditioned output in either arm exceeds what the recipe family can express, while at the registered primary 10^{-6} eighteen of arm MU's 88 valid rows sit above the family argmax, every one of them failing the tighter gate (§5.1). So the distribution the paper could not reach at submission has now been measured, and the dissolution result survives it while the ceiling result extends into it at the tighter tolerance. What remains untested is scale — arm L runs a population of five over five generations at one tier, which is a loop, not the loops of §6.

A mechanism-free alternative, now directly probed. Under an explicit "construct the packing by reasoning alone" constraint, a $k \times k$ grid at $r = 1/(2k)$ is close to the only construction whose coordinates can be emitted from mental arithmetic without error accumulation, and round($\sqrt{N}$) is the arithmetically nearest such grid. That hypothesis predicts every observation in §2–§3 without appeal to stored templates, and predicts §4.2's inversion too: the more ambitious tier attempts constructions whose coordinates it cannot compute by hand and overlaps — precisely the observed failure mode (24 overlaps and 2 nonpositive radii out of 26 `opus_alias` failures). The separating experiment — hand the model the recipe family explicitly, state $V(k, m)$ and the admissible k, and ask it to *choose* k; if it picks the argmax, the tractability reading stands — has now been run as the preregistered arm CH (§4.1). The outcome splits the hypothesis in two: the model's *choice* follows the stated score table (so tractability does not explain the selection), but its *execution* fails off-template in 30 of the 31 invalid attempts, on §4.1's disclosed post-hoc attempt-level reading (so tractability does explain the instantiation — the grid is what survives mental arithmetic). §1's vocabulary remains behavioral and no memorization claim is made. A second mechanism-level account, typicality bias in preference-tuned models (Zhang et al., 2026a), predicts concentration on the most typical construction without reference to geometry; it is not excluded by any observation here and was not tested against the tractability reading.

Contamination is probed at one level only. $N = 26$ is the canonical published cell and plausibly in training data; membership and extraction probes of the kind used to detect benchmark contamination (Carlini et al., 2021; Golchin & Surdeanu, 2024) were not run. Arm CN (Appendix J.6) answers the comparison against N values absent from the literature: at five held-out N the registered value is modal at four cells and the template structure at all five, against a registered recall alternative. The perturbed-container test an earlier draft left unrun has now been registered and run, alongside a direct-recall probe (Appendix J.7; a planted-canary test remains inapplicable to a benchmark we did not author): the perturbed-container arm CP moves the task to $[3, 5]^2$, removing the benchmark's phrasing and magnitudes, and the mapped prediction stays modal at four of five discriminating cells; the direct-recall arm RP asks for the published values outright and receives UNKNOWN in 87 of 87 scored responses, 0 of 44 recalls at the cells where a value exists to recall. Absence of the fitted cells from the pretraining corpus is not established by any of this — arm RP bounds the simplest recall mechanism (stating the value on request), not exposure.

Vendor scope. All tiers in the main study come from one provider; §5.4 moves the pooled-level claim across vendors, and its verdicts, decompositions and serving-path accountings are reported there rather than restated here. One finding from the chain's history belongs in a limitations section rather than a results one: a cross-vendor null under a fixed output budget is not evidence about the model until the budget's interaction with the vendor's response structure is checked. Arm GM2 returned 0 of 140 parseable at a 4,096-token budget; the same model at 16,384 tokens parses at 123 of 140, because the vendor API returns deliberation on a separate channel that the smaller budget died inside. Arms GM and GM3 also each required a registered serving-path amendment, both logged per row and both accounted twice (§5.4, Table 3). Scope after §5.4: the anchoring law is cross-vendor at the pooled level, it does not separate from Appendix J.2's template-shape null at the cells that can discriminate, and its full cell-level mode structure remains confirmed only within the original family.

Two instruments, and the contrast that crossed them. The agent runtime does not expose decoding parameters, so every runtime row is a distributional claim over an observed sampling regime; the pinned rows (arms P, CL, CCP and CL-W) exist to bound that, and arm P shows the two paths are not interchangeable for the weak tier (§5.3). Table 2 therefore never pools across instruments. The comparison the headline once leaned on did cross them: the Sonnet math-only cell (0 of 37, arm CCS) ran on the runtime and the Sonnet library cell (23 of 25, arm CL) on the pinned path, and the two also differ in execution budget: 10 seconds for arms CC, CC2 and CCS, 120 seconds for arm CL, a 12× gap registered on purpose (a 10-second gate measures the timeout, not the search) that the search programs the Sonnet tier writes can nonetheless be bound by. Within the weak tier the same contrast is same-instrument (0 of 12 and 0 of 11, both pinned) but arm P's code cells were executed exactly as arm CC's, at 10 seconds, so it crosses the budget too. The arm that isolates the library at the Sonnet tier, arm CC's prompt on the pinned path with the 120-second budget, was registered after those reads and ran (arm CCP, §3.5): 0 of 35 clear, none at the argmax, so the headline attributes the Sonnet gap to the library alone. What remains crossed is the weak tier's contrast, whose math-only cell ran at 10 seconds; no weak-tier program in any arm has passed the argmax under the registered parser, at either budget; the lenient reading of the library arms finds 4 of 41 (§3.5).

`opus_alias` provenance. The label denotes the serving alias addressed, not an identified model version. Without pinned weights we cannot separate a model-tier effect from a serving-path effect, and the supporting anomalies live only in the runtime transcript.

Confirmatory base for the branch rule, updated by arm M. floor and round disagree exactly on $N \in [k^2 - k + 1, k^2 - 1]$, so every discriminating cell is a trap cell by construction. The truncate arm now has five confirmed k in the square (4, 5, 6, 7, 8 — the last added by arm M's $N = 57$ cell, Appendix J.4) plus two rectangle cells; the extend arm is confirmed only at $m \leq 1$ and **disconfirmed at** $m \geq 4$ (Appendix J.4), which is the sharper scope the abstract now carries. The empirical-k back-out has been run twice as disclosed post-hoc diagnostics: `diagnostics_kmatch.py` (Appendix J.2 — 50/50 on-prediction samples k^*-structured, 64/69 valid overall) and arm M's per-cell k_{emp} distributions (Appendix J.4 — the converge cells sit on $k^*{+}1$, which is how the filler-branch failure was diagnosed).

One prompt, one phrasing — now bounded by arm PP. Every arm before Appendix J.7's paraphrase probe conditions on a single hash-locked prompt; locking buys reproducibility, not robustness to rewording. Arm PP samples three semantic-preserving paraphrases at the two discriminating cells and the registered prediction stays modal at all six paraphrase-cells (Appendix J.7), so the anchor is not keyed to the literal template string at those cells; a formatspread-style sweep of the size reported by Sclar et al. (2024) remains untested, and every other arm's claim still attaches to its own hashed prompt. Relatedly, the cross-vendor concentration differences in §5.4 are measured but not explained; alignment-induced diversity loss (Kirk et al., 2024) predicts vendor-dependent concentration and is a candidate account we did not test.

8 Reproducibility, deviations, and disclosures

Provenance in brief; Appendix I carries it in full:

- **Registration and provenance.** Every arm's prompts were hashed and its predictions and falsifiers fixed before its sampling, and each preregistration commit is a git ancestor of the sampling it governs;

no arm was selected for inclusion on its result and every arm is reported. Appendix I lists each registration, its commit, its ledgers and analysis scripts, the four limits on what the prompt digests lock, a metadata correction to the shipped ledger, the excluded records, and the internal adversarial check that preceded submission. Deviations are tabled in Appendix B.

- **Use of AI systems.** Claude models were used to draft manuscript prose and to assist implementation; the human author designed the study, set the research questions, specified the instruments and their decision rules, approved each preregistration before sampling, made the final inclusion, stopping and submission decisions, and is solely responsible for the content. The claim a reader can check is the ordering, not the authorship: each preregistration commit is a git ancestor of the sampling it governs. Because the drafting models share a family with arms under study (§3.3), what a reader is asked to trust is the released artifact — the scripts, raw ledgers, LP oracle and frozen outputs, all of which accompany this submission as anonymized supplementary material — not the author.

- **Artifact availability.** The complete artifact — raw ledgers, analysis and reproduction scripts, LP oracle, every preregistration and amendment, frozen reports, corrections ledger — accompanies this submission as anonymized supplementary material; every number regenerates via `HOW_TO_RUN.md` from the shipped ledgers. Only the pinned-path arms (P, CL) can be re-sampled by a reader from what is released: the agent-runtime rows depend on a dispatch wrapper and inherited context the digests do not lock (§3.1), so a reader can re-score every runtime row but cannot re-draw it. The working repository is a public remote whose hashes are quoted for ancestry checking; unlinked here for anonymity, linked in any camera-ready.

- **Which arms are in the body.** Every registered arm has a row in Table 1 and a ledger in Appendix I. An arm has a body subsection of its own when its registered verdict rests on discriminating cells above its own evaluability floor. An arm whose registered verdict is carried by non-discriminating cells, by fewer valid samples than its floor, or by a pass that does not separate from a template-shape null is reported in full in an appendix (arms T, G, O, GM, GM2, V, CP, RP, PP), and its body use is limited to a discriminating-cell finding or a pooled-verdict row: arm V's three evaluable aliases in §5.4, arm O's row in Table 6. Arm GM3 is the split case — its registered pooled pass rides on non-discriminating cells and is appendix-only, while its discriminating-cell finding carries a body paragraph. The criterion was applied once, in this revision, and it moved no number.

- **Stopping rule.** The manuscript closed after implementing the internal-check findings; no further data-dependent analysis was performed. Arms P and CL (§5.3, §3.5) were registered together after an external review, at corpus commit `aa473ad`, pushed before either was sampled, and each was sampled once and scored once; arm P's two post-hoc diagnostics are labelled as such in their files and in Table 3. Arms CC, CC2 and CCS (§3.4), CN (Appendix J.6), and CP, RP (with its control amendment) and PP (Appendix J.7), and L (§5.2, registered at `c9645fe` and scored once at `750bb32`) were each registered, run and scored once under their own preregistrations after that close; each analysis script was committed before its sampling and not modified after. Analyses named but not run — faithfulness split by on- versus off-prediction, interleaved re-collection of the arm-T pair, and P5 at $n = 20$ — are recorded in Appendix E.5 and §7 as future work, so that the reported statistics remain the ones the preregistrations governed.

References

Sahar Abdelnabi and Ahmed Salem. The hawthorne effect in reasoning models: Evaluating and steering test awareness, 2025. URL `https://arxiv.org/abs/2505.14617`.

Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful, 2026. URL `https://arxiv.org/abs/2503.08679`.

Mislav Balunović, Jasper Dekoninck, Nikola Jovanović, Ivo Petrov, and Martin Vechev. Mathconstruct: Challenging llm reasoning with constructive proofs, 2025. URL `https://arxiv.org/abs/2502.10197`.

Timo Berthold, Dominik Kamp, Gioni Mexi, Sebastian Pokutta, and Imre Polik. Out-of-the-box global optimization for packing problems: New models and improved solutions, 2026. URL https://arxiv.org/abs/2605.04850.

Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. Reasoning theater: Disentangling model beliefs from chain-of-thought, 2026. URL https://arxiv.org/abs/2603.05488.

Herbie Bradley, Andrew Dai, Hannah Teufel, Jenny Zhang, Koen Oostermeijer, Marco Bellagente, Jeff Clune, Kenneth Stanley, Grégory Schott, and Joel Lehman. Quality-diversity through ai feedback, 2023. URL https://arxiv.org/abs/2310.13032.

Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium*, 2021.

Mert Cemri, Shubham Agrawal, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei Zaharia, Alex Dimakis, and Ion Stoica. Adaevolve: Adaptive llm driven zeroth-order optimization, 2026. URL https://arxiv.org/abs/2602.20133.

Ziyu Chen, Yilun Zhao, and Arman Cohan. Measuring the gap between human and llm research ideas, 2026. URL https://arxiv.org/abs/2607.01233.

Erich Friedman. Packing center: circles in squares, maximizing the sum of radii. https://erich-friedman.github.io/packing/cirRsqu/, n.d. Best-known values transcribed into n_sweep_forecast.py; accessed 2026.

Yonatan Gideoni, Sebastian Risi, and Yarin Gal. Simple baselines are competitive with code evolution, 2026. URL https://arxiv.org/abs/2602.16805.

Shahriar Golchin and Mihai Surdeanu. Time travel in LLMs: Tracing data contamination in large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2308.08493.

Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.08532.

Can Gurkan, Forrest Stonedahl, and Uri Wilensky. Mutation without variation: Convergence dynamics in llm-driven program evolution, 2026. URL https://arxiv.org/abs/2606.05408.

Julien Herrmann and Guillaume Pallez. An in-depth study of llm contributions to the bin packing problem, 2026. URL https://arxiv.org/abs/2510.27353.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL https://arxiv.org/abs/2203.15556.

Yiming Huang, Biquan Bie, Zuqiu Na, Weilin Ruan, Songxin Lei, Yutao Yue, and Xinlei He. Understanding the anchoring effect of llm with synthetic data: Existence, mechanism, and potential mitigations, 2026. URL https://arxiv.org/abs/2505.15392.

Haozhe Jia. Hindsightbench: A black-box behavioral audit protocol for parametric hindsight in time-indexed llm decision tasks, 2026. URL https://arxiv.org/abs/2607.18867.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond), 2025. URL https://arxiv.org/abs/2510.22954.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL https://arxiv.org/abs/2001.08361.

Sourena Khanzadeh. Project ariadne: A structural causal framework for auditing faithfulness in llm agents, 2026. URL https://arxiv.org/abs/2601.02314.

Valentin Khrulkov, Andrey Galichin, Denis Bashkirov, Dmitry Vinichenko, Oleg Travkin, Roman Alferov, Andrey Kuznetsov, and Ivan Oseledets. Gigaevo: An open source optimization framework powered by llms and evolution algorithms, 2025. URL https://arxiv.org/abs/2511.17592.

Jinwoong Kim, Rui Yang, and Huishuai Zhang. Geobuildbench: A benchmark for interactive and executable geometry construction from natural language, 2026. URL https://arxiv.org/abs/2605.13167.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06452.

Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. Shinkaevolve: Towards open-ended and sample-efficient program evolution, 2025. URL https://arxiv.org/abs/2509.19349.

Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

Joel Lehman, Jonathan Gordon, Shawn Jain, Kamal Ndousse, Cathy Yeh, and Kenneth O. Stanley. Evolution through large models, 2022. URL https://arxiv.org/abs/2206.08896.

Pan Li. Dictionaries, not darwin: Set-level selection beats llm evolution in scientific equation discovery, 2026. URL https://arxiv.org/abs/2607.04108.

Yubo Li, Ramayya Krishnan, and Rema Padman. The chain holds, the answer folds: Trace-answer dissociation in reasoning models under adversarial pressure, 2026. URL https://arxiv.org/abs/2605.29087.

Fei Liu, Xialiang Tong, Mingxuan Yuan, Xi Lin, Fu Luo, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Evolution of heuristics: Towards efficient automatic algorithm design using large language model. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2401.02051.

Jiaxu Lou and Yifan Sun. Anchoring bias in large language models: An experimental study, 2024. URL https://arxiv.org/abs/2412.06593.

Simon Malberg, Roman Poletukhin, Carolin M. Schuster, and Georg Groh. A comprehensive evaluation of cognitive biases in llms, 2025. URL https://arxiv.org/abs/2410.15413.

Elliot Meyerson, Mark J. Nelson, Herbie Bradley, Adam Gaier, Arash Moradi, Amy K. Hoover, and Joel Lehman. Language model crossover: Variation through few-shot prompting, 2024. URL https://arxiv.org/abs/2302.12170.

Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

Alexander Novikov, Ngân Vũ, Marvin Eisenberger, et al. Alphaevolve: A coding agent for scientific and algorithmic discovery, 2025. URL https://arxiv.org/abs/2506.13131.

OpenAI. OpenAI o3 and o4-mini system card. https://openai.com/index/o3-o4-mini-system-card/, 2025.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.

Daniel Scalena, Sara Candussio, Luca Bortolussi, Elisabetta Fersini, Malvina Nissim, and Gabriele Sarti. Beyond the commitment boundary: Probing epiphenomenal chain-of-thought in large reasoning models, 2026. URL https://arxiv.org/abs/2606.13603.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11324.

Asankhaya Sharma. Openevolve: an open-source implementation of AlphaEvolve. GitHub repository, https://github.com/codelion/openevolve, 2025. Repository unpinned; circle-packing example value read at time of access, 2026.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity, 2025. URL https://arxiv.org/abs/2506.06941.

Kevin Sim, Quentin Renau, and Emma Hart. Beyond the hype: Benchmarking llm-evolved heuristics for bin packing, 2025. URL https://arxiv.org/abs/2501.11411.

Charlie Snell, Eric Wallace, Dan Klein, and Sergey Levine. Predicting emergent capabilities by finetuning, 2024. URL https://arxiv.org/abs/2411.16035.

Chang Su, Zhongkai Hao, Zhizhou Zhang, Zeyu Xia, Youjia Wu, Hang Su, and Jun Zhu. Helix: Evolutionary reinforcement learning for open-ended scientific problem solving, 2026. URL https://arxiv.org/abs/2603.07642.

Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models, 2024. URL https://arxiv.org/abs/2408.02442.

Maria Thomas, Kristina Gligoric, and Nihar B. Shah. Mitigating llm-based p-hacking by preregistering for the next llm, 2026. URL https://arxiv.org/abs/2606.27687.

Michelle Vaccaro. Preregistration for experiments with ai agents, 2026. URL https://arxiv.org/abs/2606.11217.

Niki van Stein and Thomas Bäck. Llamea: A large language model evolutionary algorithm for automatically generating metaheuristics, 2025. URL https://arxiv.org/abs/2405.20132.

Haorui Wang, Parshin Shojaee, Kazem Meidani, Kunyang Sun, José Miguel Hernández-Lobato, Teresa Head-Gordon, Jiajun He, Chandan K. Reddy, Chao Zhang, and Yuanqi Du. Towards diverse scientific hypothesis search with large language models, 2026. URL https://arxiv.org/abs/2606.10587.

Yiping Wang, Shao-Rong Su, Zhiyuan Zeng, Eva Xu, Liliang Ren, Xinyu Yang, Zeyi Huang, Xuehai He, Luyao Ma, Baolin Peng, Hao Cheng, Pengcheng He, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Thetaevolve: Test-time learning on open problems, 2025. URL https://arxiv.org/abs/2511.23473.

Marcus Williams, Hannah Sheahan, Cameron Raymond, Tomek Korbak, Deng Pan, Peilin Yang, Leon Maksin, Ningyi Xie, Phillip Guo, Ian Kivlichan, and Micah Carroll. Predicting llm safety before release by simulating deployment, 2026. URL https://arxiv.org/abs/2607.07184.

Yang Wu, Yifan Zhang, Yiwei Wang, Yujun Cai, Yurong Wu, Yuran Wang, Ning Xu, and Jian Cheng. Answer-centric or reasoning-driven? uncovering the latent memory anchor in llms, 2025. URL https://arxiv.org/abs/2506.17630.

Xia Yang, Xuanyi Zhang, Hao Hu, and Feng Ji. Beyond accuracy: Evaluating strategy diversity in llm mathematical reasoning, 2026. URL https://arxiv.org/abs/2605.09292.

Longfei Yun, Chenyang An, Zilong Wang, Letian Peng, and Jingbo Shang. The price of format: Diversity collapse in llms, 2025. URL `https://arxiv.org/abs/2505.18949`.

Dong Zhang. Testing frontier large language models' physics literacy in parallel physical worlds, 2026. URL `https://arxiv.org/abs/2607.00276`.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity, 2026a. URL `https://arxiv.org/abs/2510.01171`.

Rui Zhang and Zhichao Lu. Rethinking code similarity for automated algorithm design with llms, 2026. URL `https://arxiv.org/abs/2603.02787`.

Rui Zhang, Fei Liu, Xi Lin, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Understanding the importance of evolutionary search in automated heuristic design with large language models, 2024. URL `https://arxiv.org/abs/2407.10873`.

Xinhao Zhang, Xi Chen, François Portet, and Maxime Peyrard. What makes an llm a good optimizer? a trajectory analysis of llm-guided evolutionary search, 2026b. URL `https://arxiv.org/abs/2604.19440`.

Lexin Zhou, Wout Schellaert, Fernando Martínez-Plumed, Yael Moros-Daval, Cèsar Ferri, and José Hernández-Orallo. Larger and more instructable language models become less reliable. *Nature*, 634(8032):61–68, 2024. doi: 10.1038/s41586-024-07930-y.

A Prompts, verbatim

A.1 Bare template (`{n}` parameterized; the $N = 13$ instance hashes to `32db485bea625ff9f39f4723ebf1a01f337559a9e2cf567fb486928f71f7f8df`):

```
Pack exactly {n} non-overlapping circles inside the unit square [0,1]x[0,1]
so that the sum of their radii is as large as possible.
MUST hold: exactly {n} circles; every circle fully inside the unit square
[0,1]x[0,1] (x-r >= 0, x+r <= 1, same for y); no two circles overlap
(distance between centers >= sum of the two radii; touching is allowed).
Do not write or execute code - construct the packing by reasoning alone.
Output ONLY the raw Python list of {n} [x, y, r] lists. No explanation, no
code fences, no other text.
```

A.2 `trace_v2` template — identical to A.1 through line 3, then:

```
First write one line beginning "METHOD:" naming the construction you used.
Do not use square brackets anywhere in that line.
After the METHOD line, output ONLY the raw Python list of {n} [x, y, r]
lists. No explanation, no code fences, no other text after the list.
```

A.3 Dispatch wrapper (both arms; *not* part of the hashed task prompt, hence an instance of the fragment-hashing limitation in §3.1): `Do not use any tools.  Your entire final message must be the answer and nothing else.`

A.4 Registered prompt hashes. bare N=13 `32db485b...`, N=17 `8437df75...`, N=21 `a415425b...`, N=31 `a664d003...`, N=35 `3b08c56e...`, N=37 `2c0a8819...`; trace_v2 N=13 `a920f1c9...`, N=21 `91205727...`, N=31 `bd490b7b....` Arm CN (held-out N): N=50 `0b858558...`, N=58 `200c0bb4...`, N=62 `b9854a84...`, N=65 `5fb5fefd...`, N=75 `a5db8fc3...`, registered at commit `69a1642`. The N=43 bare hash `1208e7d2...` was ledger-only until arm P registered it (§5.3), and the pilot prompt was never hashed (§8).

A.5 Parser. Completions are parsed with `ast.literal_eval` after stripping code fences; a completion failing to yield a list of N numeric triples is recorded `literal_eval_failed` and scored invalid, with the failure mode logged (Appendix E.2 reports parse and geometric failures separately). Radii must be strictly positive; containment and pairwise non-overlap are checked at the two registered tolerances.

B Deviations and disclosure table

Table 3: Deviations from preregistration.

Prediction	Registered rule	As registered?	Outcome / reason
P1–P3	exact-value point predictions	yes	Appendix J.3, Table 4
P4 (N=37)	converge, 3.0345178	yes	confirmed
P5 (N=35)	truncate, 2.9166667, structural separation	yes	3/4 valid; $n = 4$, "consistent with"
P-S1…P-S4	directional tier comparisons	yes	§3.3
P-O1, P-O2, P-O4	tier comparisons on on-prediction / rival rates	**no**	declared not evaluable post hoc after a 13% validity collapse; the registered disconfirmation (regression toward the trap) did not occur
P-O3	multi-radius fraction ≥ 0.9 of valid	yes	trivially satisfied on 4/4 valid
P-T1	direction at each N **and** pooled $p < 0.05$	yes	not confirmed, $p = 0.30$
P-T2	directional, no alpha	yes	confirmed as registered (1/53 vs 2/50); no inferential weight claimed
P-T3	directional, no alpha	yes	met as registered; $p = 0.0325$ uncorrected fails Holm; carried by $N = 13$ (Appendix E.4)
P-T4	$\geq 90\%$ of scoreable claims match	yes, plus corrected scorer	38/41 (93%) original; 54/56 (96.4%) blind, frozen rubric
P-M1–P-M3	modal output $= V(k^*, m)$ at $N =$ 20/30/41	yes	**disconfirmed at all three cells** — modal outputs are $(k^*{+}1)$-grid values (Appendix J.4)
P-M4	$N = 57$ modal $= T(8, 57)$, rival count 0	yes	confirmed (6/10 on-prediction, 0/10 rival)
F-M1 (arm M falsifier, filler branch)	modal $\neq V(k^*, m)$ at 2+ of 3 converge cells, ties count as NOT	yes, tie-stated in registration	**triggered at 3 of 3**; $V(k, m)$ restricted to $m \leq 1$ (Appendix J.4)
F-M2 (arm M falsifier, truncate branch)	$N = 57$ modal $\neq T(8, 57)$	yes	not triggered
Arm M rejection rule	runtime deaths excluded, counted	yes (registered for this arm)	3 $N = 41$ invocations excluded, cell reported as 12 of 15
GM3 serving path	first-party API only	amendment registered **before resumption** (`arm_gm3_amendment1_openrouter.md`, `67cc4a1`)	41/140 rows via routed path, provider logged; 59.0% / 57.5% dual accounting, no verdict change (§5.4)
Arm V proposer set	3 opencode aliases	expanded 3 → 13 by amendments 1–2, each registered **before the sampling it governs**	verdicts in §5.4; `big-pickle` disclosed as vendor-unattested in amendment 1
Arm V transport repair	single pass, `max_tokens` 8192	amendment 3 registered **before resumption** (mechanical: 24,576 budget, retry, censored-row accounting)	877 attempt rows over the 325 registered slots (314 carry rows); floor evaluated per slot (§5.4)
GM3 rival-structure check	—	**post hoc**, disclosed (`diagnostics_gm3_rival.py`)	27/27 rival-valued discriminating-cell packings carry the rival's two-radius structure (§5.4); the registered signature is scoped to k^* and cannot see $(k^*{-}1)$ structure
Falsifier	trace__v2 validity `<=` bare at 2+ of 3 N	**both readings reported**	**triggered** under the registered tie-inclusive reading (2/3 cells); not triggered under the strict `<` the code implemented (Appendix E.3)
Ladder inclusion	—	**post hoc**	`opus_alias` initially excluded, later included after results known; ladder reported both ways (§4.2)
Cap exclusion	—	**not registered**	five records excluded; both rates reported (Appendix J.3)
Arm CN ledger transcription	verbatim rows	**disclosed** (`arm_cn_results.txt`)	eleven $N = 65$ rows reconstructed from the filler triple, flagged per row; discarded, P-CN1 reads 3 of 4 (Appendix J.6)
Arm M out-of-family sweep	—	**post hoc**, not preregistered	three converge-cell rows ($N = 20$ twice, $N = 30$) above the family best by 1.5–2.0 × 10^{-2}; not modal (Appendix J.4)
Arm CC2 tool use	no tools (wrapper A.3)	**disclosed** (`arm_cc2_results.txt`)	one $N = 13$ invocation used tools; scored as emitted (arm-M convention)

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
Uniform-template null	—	**post hoc**, not preregistered	discriminating-cell rates exceed the 1/3 null; binomial tails 0.043 to 2.9×10^{-4} (Appendix J.2) (`diagnostics_template_null.py`)
Arm CP (perturbed container)	P-CP1/P-CP2 + S-CP1/S-CP2 + F-CP1	registered **before sampling** (`105e8b7`); drafted at review stage, disclosed	P-CP1 **confirmed** (modal 4/5, $N = 75$ a registered tie against); S-CP1 **not met** (4/61 rival); S-CP2 5/5; F-CP1 not triggered (Appendix J.7)
Arm RP (direct recall)	P-RP1/P-RP2 + F-RP1	registered **before sampling** (`105e8b7`); draft cell N=32 → N=30, documented amendment 1	P-RP1 **confirmed**: 0/44 recalls, 87/87 scored responses UNKNOWN; F-RP1 not triggered (Appendix J.7)
Arm RP positive control	C-RP1 vs C-RP2	registered **before sampling** (`15555eb`), responding to a review finding	C-RP1: 14/15 state 0.5 at N=1; abstention-prior alternative rejected (Appendix J.7)
Arm PP (paraphrase)	P-PP1/P-PP2 + S-PP1/S-PP2	registered **before sampling** (`2e89614`); paraphrases authored results-aware, disclosed	P-PP1 **confirmed** 6/6 modal; S-PP1 **not met** (6/75 rival); S-PP2 5/6 (Appendix J.7)
Arm L (iterated loop)	L1–L4	registered **before sampling** (`c9645fe`), scored once (`750bb32`); one pooled bar per cell rather than four independent bars, a design choice that cost the arm power and is not repaired after the fact	L2 **confirmed**; L1 **not met** on 13-DIVERSE, a 2–2 tie at $n = 4$ decided against the prediction by the registered tie rule; L3 **not met and reversed**, the prediction withdrawn rather than reframed; L4 open and answered — 0 of 49 valid conditioned outputs clear the family argmax (§5.2)
Arm P (pinned path)	P-P1–P-P4 + F-P1	registered **before sampling** (`aa473ad`); post-review motivation disclosed	validity collapsed to 0/105 at the square cells; direct-emission predictions **not evaluable** under the registered floor; P-P3 holds (0/12); F-P1 denominator empty, reported as such (§5.3)
Arm P temperature diagnostic	—	**post hoc**, not preregistered	0/6 valid at temperature 0.0, 0.5 and 1.0; temperature excluded as cause (§5.3)
Arm P runtime control	—	**post hoc**, not preregistered	same prompt through the agent runtime, same day: 6/6 valid, anchor 4/6 (§5.3)
Arm P-D (runtime component)	P-PD1/P-PD2/P-PD3 + S-PD1	registered **before sampling** (`5554cc0`); run interrupted by credit at 27 of 30 rows, resume rule stated in the registration	P-PD1 **holds** and the last three rows cannot change it: thinking on 12/14 valid, modal $T(4, 13)$; system line 0/13; S-PD1 met (§5.3)
Arm CL (library code)	P-CL1/P-CL2 + F-CL1	registered **before sampling** (`aa473ad`); one execution-mechanism correction made and disclosed in the same commit	weak tier P-CL1 holds (0/11); Sonnet tier P-CL2 holds, **F-CL1 fired** (23/25, 3/3 cells); registered integration rule applied (§3.5)
Arm CL lenient reparse	—	**post hoc**, not preregistered	the 22 weak-tier parse failures are all `numpy` scalar reprs; stripping the wrapper recovers 9 valid programs (3, 4, 2 by cell) and 1 clearance at $N = 13$ (1.812531608); registered 0 of 11 stands, 1 of 20 reported beside it (§3.5) (`cl_lenient_reparse.py`)

continued on next page

Prediction	Registered rule	As registered?	Outcome / reason
Arm CCP (isolating arm)	P-CCP1/P-CCP2 + F-CCP1	registered **before sampling** (`ebf213e`); post-review motivation disclosed; one 402 interruption at row 24, resumed with one worker, request unchanged (`32fd3c7`)	P-CCP1 **holds**: 0/35 clear, none at the argmax; F-CCP1 not triggered; registered integration rule applied (§3.5)
Arm CL-W (weak-tier top-up)	P-CLW1/P-CLW2 + F-CLW1 + S-CLW1	registered **before sampling** (`f73222a`); pooling rule and the lenient secondary reading fixed in advance; 45 rather than the drafted 90, credit-limited, disclosed	P-CLW1 **holds**: 0/19 pooled clear (Wilson upper 16.8%); S-CLW1 4/41 lenient, dead zone; F-CLW1 not triggered (§3.5)
Wave stratification of the bare arm	—	**post hoc**, not preregistered	the trace effect is carried by $N = 13$ and confounded with collection wave; a diagnosis, no confirmatory weight (Appendix E.4)
Arm MU ceiling reread	—	**post hoc**, not preregistered	the clearance question was never registered for arm MU; 63 valid at 10^{-9}, none above the argmax (§5.1) (`arm_mu_ceiling.py`)
Arm CH attempt-level reading	—	**post hoc**, not preregistered	36/45 reach for the argmax against the bare arm's 5/57; the registered per-cell metric is reported first (§4.1) (`arm_f_attempt_uncond.py`)
Structural k back-out	—	**post hoc**, not preregistered	50/50 on-prediction samples k^*-structured, 64/69 overall (Appendix J.2) (`diagnostics_kmatch.py`)
Ambition diagnostic	—	**post hoc**, not preregistered	distinct radii and off-lattice fraction rise monotonically across tiers (§4.2) (`diagnostics_ambition.py`)

C The twelve exclusions of the original faithfulness scorer

Moved here from Appendix E.5, which states the consequence. The automated scorer's twelve exclusions were not claims lacking numeric content. Nine were regex coverage gaps: `DIMS_RE` matched only the literal `NxN` form, and `ROWSCOLS_RE` required rows before columns, so the following fell through despite carrying explicit checkable dimensions.

- "Regular 3 by 4 grid with one additional circle in row 4"
- "3 complete rows of 4 circles and 1 centered circle in the top row"
- "rectangular grid 4-4-4-1 with uniform radius 1/8"
- "Four-row grid with 4 circles in the first row and 3 in each of the remaining"
- "Rectangular grid arrangement with 5 columns and 4 rows, plus one additional circle on top edge"
- "Square grid 5 columns 4 rows with 1 additional circle at top"
- "Rectangular grid six columns by six rows radius one-twelfth with five circles removed"
- "Regular 5-column by 6-row grid plus one additional circle in the right margin"
- "Rectangular grid packing with five columns and six rows plus one additional circle"

Three further exclusions are closer to genuinely unscoreable: "Uniform horizontal strips with tight row spacing"; "Regular rectangular grid packing with corner circle"; "Equal-radius rectangular grid with 4 rows stacked vertically".

The nine gap cases are non-canonical phrasings, which is where a description is a priori *more* likely to diverge from the artifact. An exclusion rule that drops exactly those is not conservative in a known direction, and that is the reason Appendix E.5 reports the blind rescore over all 60 rows rather than the original 38 of 41.

D Supplementary: rectangle transfer

D.1 Transfer of the rule

For a $1 \times a$ rectangle the template has two parameters: $q^* = \text{round}(\sqrt{N/a})$ columns across the width, $p^* = \text{round}(\sqrt{N \cdot a})$ rows across the height. At $a = 1$ both collapse to $\text{round}(\sqrt{N})$ — the same rule with the aspect ratio put back, not a new rule fitted to new data, and no rectangle model output existed when it was written. It was verified against an independent LP at 213 configurations, drift below 10^{-9}, including a cross-domain check: at $a = 1$ with $p = q = k$ the square file's closed form must equal this file's LP. Shape mismatch grows as a leaves 1 — over $N = 10 \ldots 45$ the count of N whose predicted shape differs from the optimal one rises from 12 at $a = 1.0$ to 23 at $a = 1.5$, 20 at $a = 2.0$ and 23 at $a = 3.0$.

We probed the two sharpest cells with sixteen invocations (eight per cell) in a container none had been given before: $N = 19$ at $a = 3$ (predicted 3.1666667, rival 3.5749194) and $N = 25$ at $a = 2$ (predicted 3.1250000, rival 3.4832492). With all sixteen scored, **5 of 11 valid proposals landed on the predicted value and 0 of 11 reached the rival**.

We characterize all eleven here. On-prediction: two at $N = 19/a = 3$ (3.1666666667 and 3.16666673, the latter agreeing to six decimals not seven) and three at $N = 25/a = 2$ (3.125 exactly). Off-prediction at $N = 25/a = 2$: two samples emitted a uniform 5×5 grid at $r = 0.1$ summing to 2.5000000 — the *square* rule with no aspect correction, $k = \text{round}(\sqrt{25}) = 5$ — plus one at 3.0 and one at 3.151875, the latter above the 3.125 prediction from outside the family and below the rival. At $N = 19/a = 3$, **two** further samples exceeded the prediction from outside the family, 3.45 and 3.5, both below the rival.

We therefore report the rectangle as **partial out-of-sample support, not confirmation**: $5/11 = 45\%$, Wilson 95% CI $[21\%, 72\%]$, and a proposer choosing uniformly among the three or so plausible template shapes would land on-prediction about a third of the time, so 5/11 does not separate cleanly from that null. Two qualifications: validity degrades in the tall container (4/8 at $a = 3$ against 7/8 at $a = 2$, three of four $a = 3$ failures being overlaps), a separate finding and a confound on that cell at $n = 8$; and the rectangle ledger carries no prompt hashes, no run dates and no proposer fields, so the arm with the strongest transfer claim has the least provenance (§8).

D.2 Negative result: the closed form does not survive the move to rectangles

The rectangle generalizes the *rule* but not the *formula*, and we keep the failure because it is what a verification gate is for. In the square every interior vertex has four identical neighbors, so one expression covers every filler. In a $1 \times a$ rectangle, with half-spacings $h_x = 1/(2q)$ and $h_y = a/(2p)$, a filler is also capped by horizontal and vertical spacing to *adjacent fillers*. Those constraints are inactive when $h_x = h_y$ — why the square case could not have revealed them — and bind only when neighboring vertices are occupied, so the cap depends on m and on which vertices a construction uses, and no expression in (p, q, m, a) reproduces it. The LP gate caught this on the first run: at $a = 1$, $p = 2$, $q = 4$, $m = 1$ the natural generalization $\texttt{rf} = \min(\texttt{diag}, h_x, h_y)$ returns **1.125** against a true **1.1545085**, capping against a neighboring filler that does not exist. We retain closed forms only where provably exact (full grid and truncated grid) and use the LP as the value oracle for the extend branch, so what Appendix D.1 tests is the LP-backed prediction.

E Supplementary: elicitation and faithfulness

This section is explicitly secondary material and the result leans negative: the registered validity prediction fails, the concentration effect survives only as a fragile directional finding, and the faithfulness audit checks description against artifact, not against process.

E.1 Design and bundling disclosure

A ten-versus-ten pilot at $N = 21$ compared the bare prompt against a variant asking the proposer to name its construction on a leading `METHOD:` line: validity rose from 7/10 to 10/10 and the higher-scoring rival

Table 4: Forecast versus outcome, both containers.

Cell	Branch	Predicted	Rival argmax	Valid / sampled	On-pred / valid	Rival
N=13 (sq)	truncate	1.6250000	1.7761424	4/5 · **18/20**	3/4 · **10/18**	0
N=17 (sq)	extend	2.0517767	†	4/5	3/4	†
N=21 (sq)	truncate	2.1000000	2.2588835	7/10 · **15/20**	4/7 · **12/15**	2
N=31 (sq)	truncate	2.5833333	2.7485281	5/5 · **17/20**	5/5 · **13/17**	0
N=35 (sq)	truncate	2.9166667	†	4/5	3/4	†
N=37 (sq)	extend	3.0345178	†	4/5	3/4	†
N=43 (sq)	truncate	3.0714286	3.2416246	7/10	6/7	0
N=19, a=3	truncate	3.1666667	3.5749194	4/8	2/4	0
N=25, a=2	truncate	3.1250000	3.4832492	7/8	3/7	0

Plain entries are the original 45-invocation bare arm (Appendix J.3's corpus); **bold** the same cells over the full 85-row bare ledger after Appendix E added 40 rows. † marks non-discriminating cells (prediction equals family argmax), excluded from rival denominators. Square discriminating cells: 18/23 and 2/23 on the original corpus; 41/57 and 2/57 on the full ledger. Rectangle: 5/11 and 0/11. Sources: `arm_f_repro.py` over `arm_f_candidates_v2.jsonl`, `arm_g_rect.py` over `arm_g_candidates.jsonl`.

disappeared, from 2 of 7 valid bare samples to 0 of 10 trace samples. Before any scaled sample was drawn we registered four predictions and an explicit falsifier in `arm_t_preregistration.txt` (sha256 `ab7900a8...`), disclosing that the pilot's trace prompt had drifted from the bare template beyond the method line — it omitted the `[0,1]x[0,1]` tokens and reworded the output-format line — so the pilot's intervention was method-line-plus-rewording, bundled. Pilot samples are never pooled with `trace_v2`.

The scaled prompt, `trace_v2`, is itself a **bundled prompt-format-and-trace-request** intervention, though a near-minimal diff: one inserted `METHOD:` line, plus `"After the METHOD line, "` prepended to the output line and `"no other text"` changed to `"no other text after the list"`. That second change is not cosmetic — Appendix E.2 shows it doing measurable work — so the measured effect is *method-line-plus-output-line-rewording*, not the method line alone: a weaker version of the confound for which the pilot was discarded, held to the same standard.

The scaled arm ran 100 new invocations, bringing both arms to 20 per cell at $N \in \{13, 21, 31\}$. The study corpus accumulates as tabulated below. Each arm is scored under its own registration; the running total is the figure quoted elsewhere in the paper, and with arms P, CL, CCP, CL-W and P-D the study corpus now stands at 1500 invocations.

arm or group	section	invocations	running total
square (bare 85, trace 70, sonnet 30, `opus_alias` 30)	§3.3, Appendix E	215	215
rectangle	Appendix D	16	231
M	Appendix J.4	57 sampled of 60 launched	288
MU and CH	§5.1, §4.1	180, none lost	468
CC, CC2 and CCS	§3.4	135, none lost	603
CN, held-out N	Appendix J.6	75	678
CP, RP and PP	Appendix J.7	270, three rejections counted	948
L, iterated loop	§5.2	120, two rejections counted	1068
P, pinned path	§5.3	225, six refused by the serving path	1293
CL, library code	§3.5	90, four refused by the serving path	1383
CCP, isolating arm	§3.5	45, one in-flight refusal retried	1428
CL-W, weak-tier top-up	§3.5	45, none lost	1473
P-D, runtime component	§5.3	27 of 30 collected (interim), three refused by credit	1500

The square row is 85 bare (45 original + 40 new), 70 trace (10 pilot + 60 trace_v2), 30 sonnet and 30 `opus_alias`. The CP/RP/PP row is 75 offset-container, 90 recall, a 15-invocation registered positive control and 90 paraphrase invocations. The cross-vendor chain sits outside this ladder because it is scored under its own registrations: 420 GM-chain invocations (§5.4, §7) and 877 arm-V attempt rows over the 325

Table 5: Paired trace grid (scaled arm only).

N	Arm	n	Valid (parse / geom. fail)	On-pred / valid	Rival
13	bare	20	18 (0 / 2)	10 (56%)	0
13	trace_v2	20	18 (0 / 2)	16 (89%)	0
21	bare	20	15 (1 / 4)	12 (80%)	2
21	trace_v2	20	18 (0 / 2)	16 (89%)	0
31	bare	20	17 (2 / 1)	13 (76%)	0
31	trace_v2	20	17 (0 / 3)	14 (82%)	1
pooled	bare	60	50 (3 / 7)	35 (70%)	2
pooled	trace_v2	60	53 (0 / 7)	46 (87%)	1

registered invocation slots, 314 of which carry rows (§5.4). One disclosure: **20 of the 60 bare samples are pre-existing arm-F rows** ($N = 13$ ids 1–5, $N = 21$ ids 1–10, $N = 31$ ids 1–5) whose results were known when P-T1–P-T3 were written; only trace_v2 is fully fresh (Appendix E.4). Scoring used `arm_f_repro.py` unchanged with the registered 2×10^{-3} window; the Fisher exact test comes from the hypergeometric tail in `arm_t_analysis.py`, checked against a reference.

E.2 Registered outcomes

Preregistered criteria are evaluated exactly as registered, and unregistered alpha thresholds are applied nowhere.

P-T1, validity: NOT confirmed. P-T1 is the one prediction that registered an alpha (“validity $\geq$ bare at EACH N, and pooled one-sided Fisher $p < 0.05$”). Direction held at all three cells, but the pooled test gives $p = 0.30$: 53/60 (88%) trace_v2 against 50/60 (83%) bare, a difference whose detection needs several hundred samples per arm at 80% power — a failure to detect, not a demonstration of no effect. The direction that did hold is not geometric: bare had **3 parse failures and 7 geometric failures**, trace_v2 **0 parse and 7 geometric**, so among parsed samples geometric validity is 50/57 (87.7%) versus 53/60 (88.3%). The P-T1 direction is format compliance, exactly what the reworded output-format line manipulates; validity is reported split by cause from here on.

P-T2, rival suppression: CONFIRMED as registered (directional, no alpha): 1 of 53 valid trace_v2 against 2 of 50 valid bare. Counts this near zero carry no inferential weight and we claim none (Fisher $p = 0.48$, carrying no decision). One trace_v2 sample at $N = 31$ hit the rival 2.7485281 to within 3.6×10^{-8} — the first weak-tier rival hit at $N = 31$ in any arm, and formerly one of the scorer’s mismatches (Appendix E.5).

P-T3, anchor concentration: met as registered (directional), inferentially fragile. Among valid samples, 46 of 53 trace_v2 (87%) landed on the registered prediction against 35 of 50 (70%) bare. The p-value it never carried is 0.0325, one-sided uncorrected (Appendix E.4).

P-T4, faithfulness: confirmed under a corrected scorer. The registered scorer gives 38 of 41 scoreable claims matching (93%) against a 90% threshold; blind hand-adjudication under a rubric frozen before rescoring gives 54 of 56 (96.4%). Appendix E.5 reports both.

Per-cell one-sided Fisher on on-prediction (pooled bars in Figure 2): $N = 13$ $p = 0.030$; $N = 21$ $p = 0.41$; $N = 31$ $p = 0.50$. Pooled $p = 0.0325$; Holm threshold over the registered family ($m = 3$ tested with p-values) is 0.0167. The pilot (10 invocations at $N = 21$, 10/10 valid, 9/10 on-prediction) is reported separately and never summed into any total. Regenerated by `arm_t_analysis.py` over `arm_f_candidates_v2.jsonl`.

E.3 The registered falsifier, both readings

The registered falsifier reads: *“if trace_v2 validity* `<=` *bare at 2+ of 3 N, the pilot effect was the bundled rewording, not the method-line request — reported as such, not reframed.”* Observed validity: $N = 13$, 18/20 both arms — **tie**; $N = 21$, trace_v2 18/20 against bare 15/20 — trace higher; $N = 31$, 17/20 both arms —

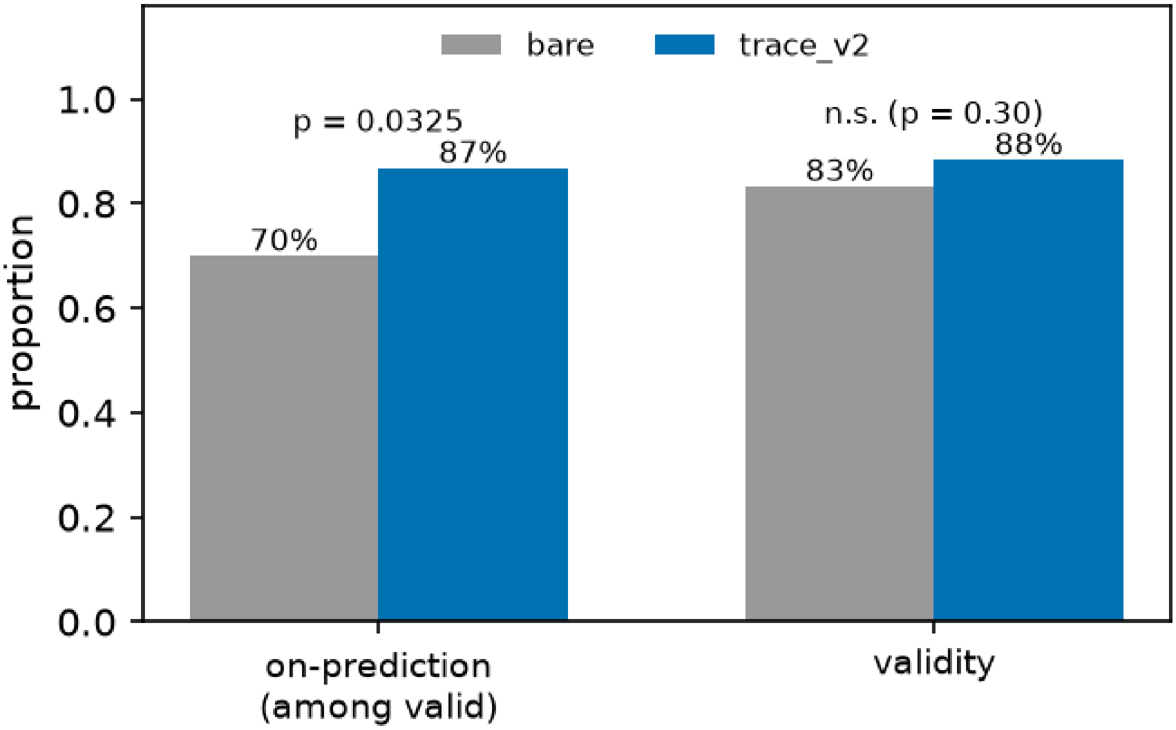

Figure 2: *Bare versus* `trace_v2`*, pooled over cells.* The annotated $p = 0.0325$ is uncorrected and fails the Holm threshold (Appendix E.4); per-cell values are in the text. Caption caveat: the bars pool the bare arm's two collection waves, which Appendix E.4 shows drift materially, so the pooled bare bar is a mixture; the same-wave comparison in Appendix E.4 is the wave-controlled picture.

tie. A tie satisfies `<=`, so under the registered wording the falsifier is **triggered at 2 of 3** N, and we report it as triggered; the code instead evaluated direction as `t_rate >= b_rate`, strict `<`, under which it fires at 0 of 3 cells. That convention appears nowhere in the preregistration, was chosen at analysis time, and read the same tie favorably twice — as "direction held" for P-T1 and as "not a falsifier cell" here.

We name the failure mode **operationalization drift between preregistration text and analysis code**: registered prose leaves boundary behavior — ties, inclusive versus exclusive bounds, rounding — implicit, and the scorer must choose after the data exist; the preregistration literature (§6) discusses what is locked, not this tie-handling layer. Remedies, free before sampling and impossible afterward: register the comparison as executable code, or state the tie convention in the registration text. `arm_t_analysis.py` now prints both readings and names the registered tie-inclusive one authoritative. Under that reading **the pilot's validity effect is attributed to the bundled rewording, not to the method-line request**, as the falsifier clause instructed; the Appendix E.2 parse-versus-geometric split points the same way, and P-T1 fails under both readings regardless.

E.4 Limits of P-T3: what the concentration result will and will not carry

P-T3 met its registered directional criterion. It is not a flagship result.

No registered alpha. P-T3 registered direction only; the p-value is a post-hoc addition.

Multiplicity. Of four registered predictions, three are reported with p-values. Holm sets the first threshold at $0.05/3 = 0.0167$; $p = 0.0325$ fails it, as it fails Bonferroni at any $m \geq 2$ and Benjamini–Hochberg FDR at $q = 0.05$. Two-sided Fisher is $p = 0.0537$. P-T3 is nominally significant uncorrected and is not under family-wise error control.

Concentration in one cell. Per-cell tests give $p = 0.030$, 0.41, 0.50 (Table 5); the pooled 17-point gap is a 33-point gap at $N = 13$ averaged with 9- and 6-point gaps elsewhere, driven by a *low bare rate at* $N = 13$ ($10/18 = 56\%$) rather than a high trace rate — the trace rate there (89%) equals that at $N = 21$. Dropping $N = 13$, the comparison is 30/35 vs 25/32, $p = 0.31$.

Wave confound. The bare arm mixes two collection waves; trace_v2 is one. Splitting bare on the registered id boundary, with no intervention applied to either half, on-prediction is: $N = 13$, 3/4 old wave vs 7/14 new; $N = 21$, 4/7 vs 8/8; $N = 31$, 5/5 vs 8/12 — the control arm's own between-wave drift is comparable to the effect attributed to the manipulation. (An earlier draft printed 2/4 vs 10/11 at $N = 21$ and a 25/37 pooled figure — arithmetically impossible against Appendix E.1's id boundary; corrected here from the ledger.) Restricting to one wave, new-wave bare is 23/34 (68%) against trace_v2's 46/53 (87%), but **this stratified comparison is post hoc, not preregistered, no confirmatory weight** — a diagnosis, not a rescue: wave is a confound *in the registered analysis*, which cannot separate it from the manipulation. The design fix, interleaved collection in one session, was not run.

What we therefore claim. A minimal method-naming request does not make the proposer better at geometry and does not measurably change which rare constructions it reaches; in the registered direction it concentrates output onto the anchor, but that concentration fails correction, is carried by one cell, is confounded with collection wave, and is bundled with an output-format rewording that demonstrably affects parse compliance. Studies collecting process descriptors *by asking for them*, including descriptor-driven quality-diversity pipelines where the descriptor is a requested self-report, should treat this as a reason to check, not a coefficient to apply. The pilot also showed a *third* effect in the P-T3 direction at the same cell (trace 9/10 on-prediction against bare $N = 21$ at 4/7, one-sided $p = 0.16$), so P-T3 replicated a pilot signal rather than testing a blind prediction — a different status, and the prereg's pilot disclosure covered the prompt drift only.

E.5 Faithfulness with ground truth

Claim and artifact sit in the same completion and the artifact is fully determined, so a method line is checkable against the emitted coordinates. The original automated scorer, `arm_f_repro.trace_faithfulness()`, returned 38 of 41 scoreable claims matching (93%) against the registered 90% threshold, but nine of its twelve exclusions were regex coverage gaps rather than claims without numeric content, and the excluded set is enriched for non-canonical phrasings where a mismatch is a priori *more* likely — so the exclusion was not conservative in a known direction. Worst case, had all twelve been unfaithful, the rate would have been $38/53 = 72\%$. Appendix C lists the twelve exclusions verbatim and names the two regexes that produced them. That accounting is why the arm was rescored blind.

Blind rescore under a frozen rubric. The rubric (`p7_faithfulness_rubric.md`) — checkable assertions in any surface form, plus match rules including a base-grid convention for "grid plus additions" claims and a truncation convention for "with k removed" claims — was committed to git as `e181d2a` at 03:56:56 on 2026-08-01, **before** any rescoring run, so the freeze is checkable, and it pre-committed the reporting rule: below 90%, P-T4 reported not confirmed, full stop, no consolation framing. Scoring was blind — claim text and circles only, no predicted or rival values, no running tally, no threshold and no sight of the existing 38/41 result — with tallying afterward; all 60 labels are released verbatim in `p7_blind_labels.json`.

Result. Over all 60 trace_v2 rows: **54 MATCH, 2 MISMATCH, 4 UNSCOREABLE** — a corrected rate of **$54/56 = 96.4\%$** of scoreable claims, Wilson 95% CI $[88\%, 99\%]$, passed, and on 56 scoreable claims rather than 41. The CI's lower bound sits just below the registered threshold, so "clears 90%" is a statement about the point estimate at this n. The two mismatches share a form: row 11 claims "alternating rows of 4 and 3 circles" over observed row sizes 4,3,3,3, row 33 claims "5 rows alternating between 5 and 4 circles" over observed 5,4,4,4,4 — both *mis-descriptions of alternation*, a repeating pattern named and built for the first period only. The original scorer's three mismatches were three different claim forms, not one; the filler-adds-rows mechanism the original scorer described was real, and the rubric's base-grid convention handles it, so all three score MATCH under hand adjudication.

Scope, three ways. The blind pass covers *invalid* rows too: the original 93% conditioned on completions that produced a correct packing — the wrong conditioning, since a method line attached to an overlapping layout is where description and artifact are likeliest to diverge — while the blind pass scored all 60 rows including the 7 invalid ones, 4 of the 60 unscoreable. Second, the check verifies the description against the *artifact*, not the process: the non-claim guard applies in full, and for a model already emitting grids, "5×5 grid" matching a 5×5 grid is closer to a self-consistency check than to the causal question the chain-of-thought literature asks (§6). Third, the audit is confounded with the elicitation arm, computed on the arm Appendix E.2 shows concentrates 87% of valid outputs onto one template, where correct description is close to free; faithfulness is informative precisely on off-template outputs, such as the pilot's triangular-hexagonal 6+5+4+3+2+1 sample, described exactly while summing to 1.75. The split by on- versus off-prediction is not run here (§8, stopping rule). What survives is narrow: in a domain where claims are checkable, the method lines this proposer emits are, at 54 of 56 scoreable claims, true of the object it produced — and both failures are a model describing a pattern it started and did not finish.

F Arm V in full: cross-vendor zero-shot under free-tier serving

Moved from §5.4 in the structural cut; the summary there is the contract, this is the report.

Arm V. Arm GM3 varied the vendor while holding the instrument close to laboratory grade — first-party API, pinned temperature, explicit output budget, seven cells at twenty samples. Arm V varies vendor and serving class at once, under the same byte-identical arm-F prompts both arms inherit (SHA-256 per N restated in the registration and recomputed at call time, aborting on mismatch; the GM-chain registration pins the identical hashes), at five cells and five samples per (model, N), against free-tier aliases across seven attributed vendors (`arm_v_preregistration.md`, committed before sampling; three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths — the opencode CLI and OpenRouter chat-completions — and raised `max_tokens` to 24,576 after hidden-reasoning truncation emptied three models' replies). Serving disclosures per the companion paper's protocol: aliases on managed paths, `sampling_params: null` on every row, per-row upstream-provider strings logged where the path reports them, every invocation appended live to `arm_v_candidates_raw.jsonl` — 877 invocations, failures and refusals included. Scoring is `arm_f_repro.py`'s parse/validate/classify unchanged, tolerances registered in arm F before its sampling.

Registered floor: $\geq$10 of a model's 25 registered invocations valid at 10^{-6}; below it, no anchoring claim in either direction. Execution retried each invocation slot until content arrived, per amendment 3's registered transport-repair and accounting rule, so the ledger holds many more attempt rows than slots — gemma-4-31b's 25 slots, for instance, took 190 attempt rows, 165 of them transport failures — and the floor is evaluated on the 25 slots, not the attempts. Ten of thirteen aliases finished BELOW-FLOOR — including all three originally-registered opencode aliases, whose logged failure taxonomies are transport-dominated for one (CLI build banners, timeouts, "Unknown model" errors as the free tier rotated its catalogue) and roughly half transport, half parse-failure for the other two — plus one alias returning all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split:

alias (vendor)	valid	V1 on-prediction	V1 verdict (bar: $\geq$50%)	V2 rival-argmax (bar: $\leq$10%)
`north-mini-code` (Cohere)	12	8/12 [39–86%]	**TRANSFERS** (point est.)	0/4 → **HOLDS**
`gpt-oss-20b` (OpenAI, open-weight)	18	11/18 [39–80%]	**TRANSFERS** (point est.)	0/9 → **HOLDS**
`gemma-4-31b-it` (Google)	15	0/15 = 0%	**DOES-NOT-TRANSFER**	0/3 → **HOLDS**

This is the registration's most informative branch pair. Anchoring under this instrument is not an artifact of the original family: two further vendors' models land the registered construction values at 61–67% of valid outputs, point estimates above the 50% bar.

Both passes are weaker than those percentages suggest, in three ways we state rather than leave to the reader. The bracketed Wilson 95% intervals include the bar, so each verdict is a point-estimate pass at $n = 12$ and $n = 18$. Three of thirteen aliases were evaluable at all; the other ten finished BELOW-FLOOR and claim nothing in either direction. And the pooled rate spans cells that cannot discriminate anchoring from family search. Decomposing onto the discriminating cells alone — the split this section demands of the GM chain, and symmetry demands here — gives 3 of 4 and 5 of 9. Unlike GM3 the decomposition does not reverse either verdict, but it shows how little of each pass sits where prediction and family argmax differ.

Appendix J.2's template-shape null belongs here too, and applying it costs the arm its strongest reading. Against that null a proposer lands on-prediction one time in three. The pooled rates clear it — exact upper tails 0.019 and 0.014 — but they clear it on cells where prediction equals the family argmax. At the cells that can discriminate, neither pass separates: 3/4 gives 0.111 and 5/9 gives 0.145, and GM3's 11/43 gives 0.895, below the null. So the honest cross-vendor statement is that two further vendors are *consistent with* the behaviour and that at $n = 4$ and $n = 9$ the discriminating evidence does not separate them from a shape-uniform proposer. Recomputed in `cross_vendor_null.py`; post hoc, like the null itself.

The strong form holds everywhere evaluable *within this arm*: zero rival-argmax emissions in 16 discriminating-cell validities. The registration (`arm_v_preregistration.md`) recorded the original family's prior for the same statistic as 0/21 over $N \in \{13, 31\}$ only, a narrower scope than Appendix J.3's 2/23 whose two rival hits both sit at $N = 21$, so the two corpora agree at 0 on the registration's own cells. The standing exception is gemma-4-26b under the GM-chain instrument, whose discriminating-cell mode *is* the rival.

The transfer's point estimates sit at or below the in-family rates: 61–67% on-prediction against 56–76% in-family at the same five cells (Appendix J.2). The ranges overlap, so reduced concentration is a directional reading these n cannot separate. Directionally lower concentration at different vendors, on byte-identical prompts, landing on the same modal value is what a shared attractor predicts; equal concentration would be harder to separate from shared training text. And it is not universal: gemma-4-31b, clearing the same floor, lands the prediction never. Its valid packings are unanchored and low (bests 0.88–1.85 against predictions 1.63–3.03) — not the rival, not the template, just weaker de-novo construction. Under the registration's falsification map, the family-specific branch fires — any model clearing the floor with DOES-NOT-TRANSFER makes the closed form a family-bounded property, stated in §3.3 with data rather than a caveat — while the map's all-transfer branch fails on gemma alone, so the reading that anchoring is an artifact of the original family is defeated by the same table that bounds its generality.

The gemma pair needs stating precisely, because this family now appears on both sides, and the prompts are identical across the two arms — what differs is parameter count, serving path and sample budget, not the elicitation. Under the GM-chain instrument (first-party, pinned temperature, 7×20), gemma-4-**26b** met the pooled bar at 57.5% while its discriminating-cell rate (26%) fell below it (the mixed-serving accounting; first-party-only, 59.0% — §7's registered deviation applies to this figure) while emitting the *rival* construction at 63% of discriminating-cell validities; under arm V's free-tier instrument (5×5), gemma-4-**31b** through a rate-starved alias (Google AI Studio upstream; 27–38 of each cell's attempts returned as shared-pool 429s before content arrived) lands neither the prediction nor the rival — its valid packings are weaker de-novo constructions. The pair bounds the family's behaviour rather than contradicting itself — one member sits *above* the template's execution reach (far enough to build the drop-and-fill rival the original family cannot), the other lands below it — but model variant and instrument quality change together across the two measurements, so which difference carries the split is not attributable from this pair; either way, no single-model measurement would have seen the boundary.

G The `opus_alias` arm in full

Moved from the body in the structural cut; the tier ladder that §4.2 quotes is Table 6.

‡ Addressed through a bare serving alias; the alias-to-weights binding is not attestable, and this row is not a statement about any released model version. Tier denominators are not matched (the weak tier spans seven N against three), which is why the matched-cell row is given separately. Sources: `arm_f_candidates_v2.jsonl`, the three preregistration files.

The top tier attempts recursive constructions and fails to build them. The third arm was invoked through a bare tier alias. We name it `opus_alias` throughout and make no claim about which weights served it: the runtime accepts only the bare alias and exposes no dated identifier, so the binding is a vendor promise rather than an attestable fact. No version number appears anywhere in this paper.

Table 6: Three-attractor ladder. Valid columns count samples passing containment and non-overlap at the 10^{-6} and 10^{-9} tolerances; On-pred and Rival count valid samples landing on the predicted value and on the rival argmax. The matched-cell row is the comparison to quote across tiers (83% → 100% → 13%).

Tier	Cells	n	Attempted family	Valid 10^{-6}	Valid 10^{-9}	On-pred	Rival	Failure mode
haiku (bare)	13, 17, 21, 31, 35, 37, 43	45	uniform grid, truncated; fillers when underfull	35 (78%)	29 (64%)	12/16 (3 matched cells)	2	overlap 6, outside 1, parse 3
haiku, matched cells	13, 21, 31	60	as above	50 (83%)	—	35/50	2	overlap 5, outside 2, parse 3
sonnet	13, 21, 31	30	perturbed grid, hexagonal rows, 2–3 radii	30 (100%)	27 (90%)	1/30	5 (structure)	none
`opus_alias` ‡	13, 21, 31	30	quarter-circle corners, Apollonius fillers, coarse grid + border strips	4 (13%)	4 (13%)	0/4	0	overlap 24, non-positive radius 2

That caveat matters: completion-time and token-count anomalies were observed in the unreleased session transcript for this arm, but the ledger carries no latency or token field, so they appear in no released artifact and nothing below relies on them.

Under those caveats the arm — preregistered fully blind in `arm_o_preregistration.txt` — was valid in 4 of 30 invocations (13%) at both tolerances. At every cell it attempted a more ambitious construction than either other tier: quarter-circle corners at $r = 0.25$ with Apollonius-style fillers at $N = 13$ (10/10 samples), mixed-radius 4×4-ish grids at $N = 21$, and a coarse 3×3 grid at $r \approx 1/6$ with border strips at $N = 31$ (0/10 valid). Failures are geometric rather than numerical — edge strips at $r = 0.03$ placed 0.138 from an $r = 1/6$ grid circle needing 0.197 — and twice the arm padded to count with zero-radius circles. Valid samples score *below* the trap they were expected to fall into (1.26–1.41 at $N = 13$ against 1.625). A post-hoc diagnostic (not preregistered; labeled as such in the repository) finds no tolerance artifact behind the 13%: 24 of 26 invalid samples overlap grossly (median maximum overlap 3.3×10^{-2}; radius-shrink repair costs a median 15% of sum-of-radii), while tolerance-scale near-misses ($< 2.5 \times 10^{-5}$) fall instead in 5 of 7 geometry-scored weak-tier failures — the exact-tangency grids, not the ambitious constructions.

Two decisions here were made after seeing the data and are recorded as deviations (Table 3). P-O1, P-O2 and P-O4 are reported **not evaluable**, on the grounds that a validity collapse of this size makes an on-prediction tier comparison uninterpretable; P-O3 is trivially satisfied on 4/4 valid samples; the registered disconfirmation — regression toward the trap — did not occur. De-registering three of four predictions in the one arm that disconfirmed is exactly the move preregistration exists to constrain, which is why it is tabled rather than narrated. And the arm was initially excluded from the ladder, then included once its results were known; we follow the later decision and report the ladder both ways (Table 6) — with the arm 83% → 100% → 13% on matched cells, without it 83% → 100% plus the qualitative shift from truncation to perturbed multi-radius hybrids, which is already the publishable contrast.

Two rungs, two corpora, and they are easy to confuse. The 83% is the 60-row matched-cell ledger at $N \in \{13, 21, 31\}$. The 78% that circulates beside it is the bare arm's own 45-row ledger over all seven weak-tier cells — a different corpus and a different cell set, not a fourth rung. The 10^{-9} ladder reads 64% → 90% → 13%, where the 64% is again the 45-row seven-cell ledger and its matched-cell counterpart was not computed, so the two ladders are not directly comparable at their first rung either. Both rows appear separately in Table 6 and neither figure is quoted without its corpus.

H Denominators: which on-prediction number is which

Moved from Appendix J.2 in the structural cut. The headline figure for the anchoring claim is 41/57; everything below is a subset, a superset, or a different question, and each is named against its own corpus.

Which on-prediction number is which. The rate is reported on several corpora and cell sets, each for a reason, and the figures are easy to mistake for disagreements. They are gathered here once, so that every later mention can be read against its own denominator.

figure	counts	corpus	cells
56–86%	on-prediction	bare weak-tier arm	each of the seven square cells
18/23	on-prediction	original corpus	the four discriminating square cells
41/57	on-prediction	full shipped ledger	the same four
2/23, 2/57	rival emissions	those two corpora	the same four
9/12	on-prediction	full shipped ledger	the three non-discriminating square cells
12/16	on-prediction	original 45-row arm	the three matched cells (13, 21, 31)
35/50	on-prediction	full matched-cell ledger	the same three
47/102	on-prediction	full ledger plus both higher tiers and the rectangle	the four discriminating square cells, plus those tiers and that container

I Provenance, in full

Section 8 summarises this; here it is in the form the body carried before the structural cut, so that a reader auditing ancestry finds every registration, commit, ledger and script in one place.

- **Registration.** Prompts were hashed with SHA-256 and the hashes recorded before sampling, with the predicted values they were to be tested against: `arm_f_repro.py` (header, P1–P5, square arm), `arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt` (prereg hash `ab7900a8...`), `arm_m_preregistration.txt` with `arm_m_prompts.json` (committed `e528c6b` before sampling), and `arm_mu_preregistration.txt` with `arm_mu_prompts.json` (committed `2b7d202` before sampling; decision thresholds, power analysis, tie conventions and falsifier F-MU1 all registered), and the cross-vendor chain `arm_gm_preregistration.txt` (commit `37b3adb`), `arm_gm2_preregistration.txt` (`3019aab`), `arm_gm3_preregistration.txt` (`c0b5b97`) with its serving-path deviation registered in `arm_gm3_amendment1_openrouter.md` (`67cc4a1`) before the affected cells were sampled, and `arm_v_preregistration.md` (commit `5444f10`, ancestry-checked by its runners before any sampling) with its three amendments, each committed before the sampling it governs. Arm CC's registration (`arm_cc_preregistration.txt` with frozen prompts, executor and smoke test) was committed and pushed to the public remote at `285deca` before its first invocation; arms CC2 and CCS were registered together (`arm_cc2_preregistration.txt`, commit `4d7b7c5`, pushed before sampling) after CC's results were known, with that motivation disclosed in the registration itself. Arm CN (`arm_cn_preregistration.txt`, commit `69a1642`, pushed before sampling) was registered after the rest of the manuscript was complete, in response to a pre-submission check; its motivation is post-hoc with respect to the paper's results and is disclosed in the registration, and its predictions were fixed before its first invocation. Arms CP and RP (`arm_cp_preregistration.txt`, `arm_rp_preregistration.txt`, each with its prompt-hash JSON and analysis script, commit `105e8b7`, pushed before the first invocation) were drafted at review stage and registered together; RP's one draft-to-registration cell change ($N = 32 \to 30$) is documented in its registration, as is its positive-control amendment (`15555eb`). Arm PP (`arm_pp_preregistration.txt`, prompt hashes and scorer committed, `2e89614`, pushed before the first invocation) discloses that its paraphrases were authored results-aware. The rectangle transfer was registered before collection and never refitted. Arms P and CL (`arm_p_preregistration.txt`, `arm_cl_preregistration.txt`, builders, frozen prompts, runners and analysis scripts) were committed and pushed together at `aa473ad` before either was sampled; ledgers `arm_p_collect.jsonl` and `arm_cl_collect.jsonl`, frozen reports `arm_p_report.json` and `arm_cl_report.json`, and the two post-hoc diagnostics `arm_p_diag_temperature.jsonl` and `arm_p_diag_runtime.jsonl`, each labelled post hoc in its header. On multiplicity: this study now spans many registered arms, and no correction across arms is applied — each arm's predictions were registered with its own bars and falsifiers before its sampling, every arm is reported regardless of outcome (three GM-chain instruments failed or stalled before one succeeded, and Appendix J.4's falsifier fired), and no arm was selected for inclusion on its result. The protection against garden-of-forking-paths here is registration and exhaustive reporting, not a family-wise error rate; a reader who wants one can count the twenty-five arms from §3's table. That defence covers the registered predictions and not the post-hoc diagnostics: which unregistered analyses we chose to run is a degree of freedom registration does not constrain. Table 3 lists them, and three cut against us — Appendix J.4's out-of-family sweep, §5.4's rival-structure check, and Appendix E.4's wave stratification.

- **Four limits on that claim.** (i) The digests cover the task prompt only; the subagent inherits instruction files outside it, so what is locked is a *fragment* of the conditioning context, a replicator cannot reconstruct those files, and we cannot establish they were identical across the arm-F and arm-T windows — the boundary Appendix E.4 identifies as a confound. (ii) `arm_f_prompts.json` covers $N = 13$, 17, 31, 35 and 37; the $N = 21$ bare hash is registered in `arm_t_preregistration.txt`; the $N = 43$ **bare hash appeared in no preregistration and no prompts file** until arm P's, existing only in the ledger — it recomputes exactly from the bare template, but recomputation after the fact is not registration. (iii) **No hash was ever registered for the pilot prompt**, whose drift the preregistration describes only in prose; the ten pilot rows carry `prompt_sha256: null` in the corrected ledger with an explanatory note. (iv) The rectangle ledger (`arm_g_candidates.jsonl`) carries no prompt hashes, no run dates and no proposer fields at all.

- **Ledger metadata correction.** Three internal-check findings established that the shipped ledger's provenance fields contradicted the paper: trace rows carried the *bare* prompt hash for their cell, `run_date` was uniformly `2026-07-30` on all 215 rows, and the Sonnet and `opus_alias` rows were stamped with the Haiku alias and dated id. All three are corrected in `arm_f_candidates_v2.jsonl`; the correction is metadata only — every scored quantity, geometry, raw output, arm label and sample id is byte-identical to v1, verified field by field, each corrected field carrying a sibling `*_v1` field. `arm_f_candidates.jsonl` is unmodified (sha256 `02ecabcd...`); v2 is `9f845f78...`. Correction rules, sources, wave-boundary arithmetic and caveats — including that `run_date` has day granularity only — are in `ledger_v2_corrections.md`. Analysis scripts read v2.

- **Excluded and unreleased records.** The five concurrency-cap rejections appear in the ledger in no form, no `excluded_reason` field exists, and the exclusion rule was not preregistered; both rates are reported in Appendix J.3. The `opus_alias` latency and token-count anomalies (§4.2) come from the runtime session transcript; no latency or token field exists in any released artifact.

- **Analysis artifacts.** Raw outputs are stored verbatim (`arm_f_raw.json`, `arm_f_candidates.jsonl`, `arm_f_candidates_v2.jsonl`, `arm_g_candidates.jsonl`, `arm_m_collect.jsonl` scored by `arm_m_analysis.py`, `arm_mu_collect.jsonl` scored by `arm_mu_analysis.py` with frozen output `arm_mu_results.txt`, and the GM3 ledger `arm_gm_gm3_checkpoint.jsonl` — every row's raw API response verbatim, serving path and provider tagged per row — scored by `arm_gm3_analysis.py` into `arm_gm3_report.json` with both serving-path accountings) and scoring is deterministic and local. Value claims are gated on an independent LP oracle rather than on the closed form being tested: `n_sweep_forecast.py` with `verify_against_lp` checks 83 configurations to within 10^{-9}, and the same oracle produced Appendix D.2's negative result. The faithfulness rubric (`p7_faithfulness_rubric.md`) was frozen by git commit `e181d2a` (2026-08-01 03:56:56) before any rescoring, and all 60 blind labels with justifications are in `p7_blind_labels.json`. `arm_t_analysis.py`, `n_sweep_forecast.json`, `rect_forecast.json`, `p11_mode_baseline.json` and `STATE.md` are included; every table regenerates from raw outputs by a command documented in `HOW_TO_RUN.md`.

- **Internal checks.** Before submission the manuscript passed an internal adversarial check: language models of other families (`deepseek-v4-pro`, `deepseek-v4-flash`, Gemini), under written protocols, independently recomputed the paper's statistics from the released ledger; thirty-two claims were corrected as a result, itemized in the supplementary `corrections_ledger.md`. This is quality assurance, not peer review, and carries no external authority.

I.1 Per-arm ledgers and scripts

The sentence each arm's subsection used to end with, verbatim, under the subsection's title.

- **Arms MU and CH: no transfer to one-parent calls; the argmax in hand does not dislodge the template** Raw rows verbatim in `arm_mu_collect.jsonl` (180 of 180 launched invocations collected across arms MU and CH, no runtime deaths); scoring in `arm_mu_analysis.py`, committed before sampling completed; frozen output in `arm_mu_results.txt`. This arm brings the study corpus to 468 invocations.

- **Arm CC: an executed program does not escape the family** Raw rows verbatim in `arm_cc_collect.jsonl` (45 of 45 launched invocations collected, no runtime deaths); scoring `arm_cc_analysis.py`; frozen report `arm_cc_report.json` and summary `arm_cc_results.txt`.

- **Arm P: the pinned path** Registration `arm_p_preregistration.txt`, prompts and hashes `arm_p_prompts.json`, runner `arm_p_run.py`, scorer `arm_p_analysis.py`, ledger `arm_p_collect.jsonl` (225 rows, six refused by the serving path), report `arm_p_report.json`; the two post-hoc diagnostics in `arm_p_diag_temperature.jsonl` and `arm_p_diag_runtime.jsonl`, labelled post hoc in their headers. Arm P-D: registration `arm_pd_preregistration.txt` (commit `5554cc0`), builder `arm_pd_build.py` (arm P's $N = 13$ prompt, hash asserted), spec `arm_pd_prompts.json`, runner `arm_pd_run.py`, scorer `arm_pd_analysis.py` (arm P's scorer imported unmodified), ledger `arm_pd_collect.jsonl` (27 rows plus the 402 rows kept for resume), report `arm_pd_report.json` marked INTERIM.

- **Arm CL: the library channel** Registration `arm_cl_preregistration.txt`, builder `arm_cl_build.py` (one clause replaced in arm CC's prompt, hash asserted), prompts `arm_cl_prompts.json`, runner `arm_cl_run.py`, scorer `arm_cl_analysis.py` (driver and `numpy`/`scipy` versions recorded in the report), ledger `arm_cl_collect.jsonl` (90 rows), report `arm_cl_report.json`. The post-hoc re-execution and lenient reparse are `recount_cl.py` and `cl_lenient_reparse.py` in the paper repository's `loop/` directory, outputs `cl_recount.json` and `cl_lenient.json`.

- **Arm CCP: the isolating arm** Registration `arm_ccp_preregistration.txt` (commit `ebf213e`), builder `arm_ccp_build.py` (arm CC's prompts byte-identical, hashes asserted), prompts `arm_ccp_prompts.json`, runner `arm_ccp_run.py` (the one-worker amendment after the row-24 interruption, `32fd3c7`), scorer `arm_ccp_analysis.py` (arm CL's pipeline with the allowlist set to `math`), ledger `arm_ccp_collect.jsonl` (45 rows plus the one 402 row, both segments timestamped), report `arm_ccp_report.json`.

- **Arm CL-W: the top-up** Registration `arm_clw_preregistration.txt` (commit `f73222a`), builder `arm_clw_build.py` (arm CL's weak-tier prompts byte-identical, hashes asserted), prompts `arm_clw_prompts.json`, runner `arm_clw_run.py`, scorer `arm_clw_analysis.py` (arm CL's pipeline over the pooled 90 rows, both readings per row), ledger `arm_clw_collect.jsonl` (45 rows), report `arm_clw_report.json` with every row's registered and lenient result.

- **Arms CC2 and CCS: the replication and the tier probe** Ledgers `arm_cc2_collect.jsonl` and `arm_ccs_collect.jsonl` (45/45 each, no runtime deaths); runner `arm_cc2_analysis.py` reuses CC's registered pipeline unmodified. These arms bring the study corpus to 603 invocations.

J The mode prediction and the arms that bound it

The closed form that names the weak tier's modal output in advance, reported in full; the body keeps only what Table 2 and §4 need from it.

J.1 Trap zones and the bound table

Substituting $k^*(N)$ into the branch condition: TRAP $N \in [k^2 - k + 1, k^2 - 1]$, CONVERGE $N \in [k^2, k^2 + k]$. Over $N = 10 \dots 60$: $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$ and $[57, 63]$ clipped to $[57, 60]$ by the sweep bound.

Inside a zone the rule generally loses value against the best construction available *within the recipe family*. Worst-in-zone penalties fall as N grows: 8.51% at $N = 13$, 7.03% at 21, 6.01% at 31, 5.25% at 43, 4.66% at 57. The penalty reaches zero at $N = 35$, 48, 14 and 15, but **not** at $N = 24$, where truncation gives $T(5, 24) = 2.4000000$ against $V(4, 8) = 2.4142136$, a residual 0.59%. Branch label and penalty are therefore not coextensive: 4 of the 22 trap-branch N in range cost nothing, so "trap" names the branch, not a guaranteed loss. Where the penalty is zero, truncation is separable from convergence only by structure; Appendix J.3 uses $N = 35$ as that control.

The linear-programming verification summarized in §2.2 originally covered $k = 2 \dots 7$ and stopped short of two registered cells, arm M's $T(8, 57)$ and arm CN's predictions at $k = 8$ and 9; the sweep was extended rather than the claim narrowed (`lp_extend_k89.py`), and every registered closed-form value is inside the verified range.

Our bound table (`n_sweep_forecast.json`, transcribed from Friedman (n.d.)) covers $N = 10 \dots 30$ only, and its $N = 26$ entry, 2.63598, matches ShinkaEvolve's figure truncated. That entry has since been re-credited on the source page to a July 2026 submission, so we withdraw the reading that LLM-driven systems hold the record here and claim only what the table supports. What survives is that the recipe family is never competitive with that record: deficit 0.02–0.26 across $N = 10 \dots 30$, and 0.0946 at $N = 26$ ($2.63598 - 2.5414214 = 0.0945586$). Above $N = 30$ there is no bound table, so both the deficit claim and the LP gate's "never exceeds a published bound" abort are unchecked there, covering three of five trap zones and four of our seven square cells.

J.2 The mode prediction: seven cells, all modal

An exact point prediction invites a fair objection: a per-sample hit rate well short of 100% is not "predicting the exact output". The number should be read against the sampling entropy of the distribution, computed in `p11_mode_baseline.json` (bare arm, 2×10^{-3} buckets). At **all seven tested N the predicted value is the empirical modal output** — four discriminating cells plus three where the prediction and the family argmax coincide, so the anchoring-specific evidence sits in the four — and the on-prediction rate equals the modal frequency to within one sample. The two columns are counted on different partitions: the modal-frequency column groups at four decimals, the on-prediction column uses the registered 2×10^{-3} window. They come apart at exactly one cell, $N = 31$, where the window admits one packing the four-decimal bucket separates out, which is why that row reads 12/17 against 13/17:

N	predicted	empirical mode	modal freq	on-prediction / valid	k^*-structure / valid†
13	1.6250000	1.6250	10/18	10/18 (56%)	17/18
17	2.0517767	2.0518	3/4	3/4 (75%)	3/4
21	2.1000000	2.1000	12/15	12/15 (80%)	13/15
31	2.5833333	2.5833	12/17	13/17 (76%)	17/17
35	2.9166667	2.9167	3/4	3/4 (75%)	3/4
37	3.0345178	3.0345	3/4	3/4 (75%)	4/4
43	3.0714286	3.0714	6/7	6/7 (86%)	7/7

† Post-hoc structural check, not preregistered, run after the square arm closed (`diagnostics_kmatch.py`): the grid order backed out of each sample's dominant radius ($k = \mathrm{round}(1/2r)$) equals $k^*(N)$. All 50 of 50

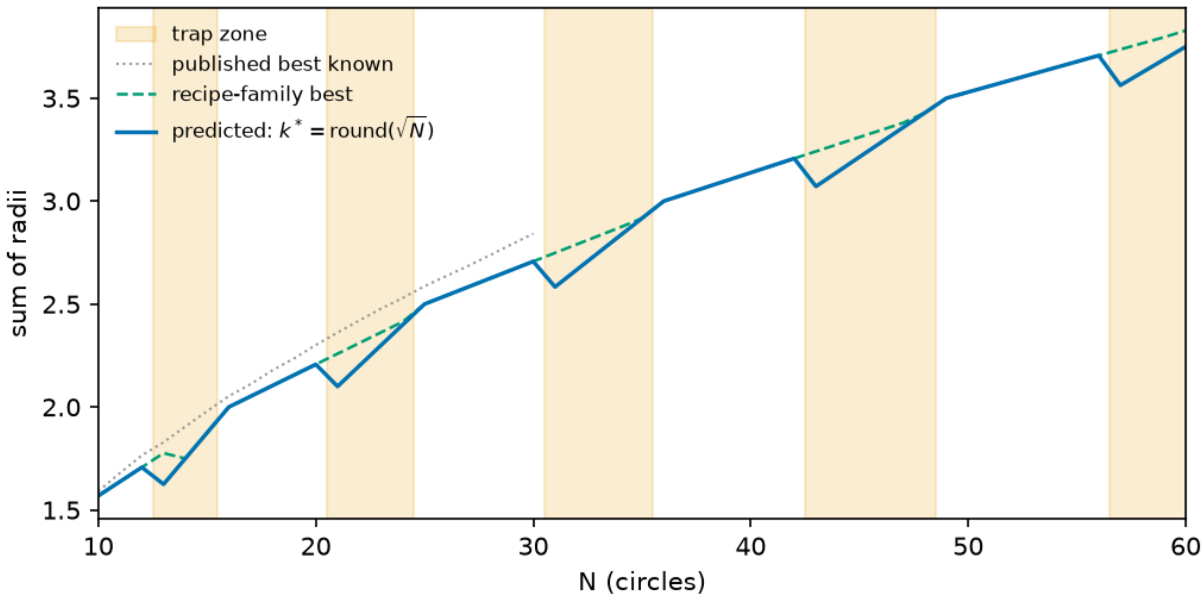

Figure 3: *Predicted versus optimal value,* $N = 10 \dots 60$. (i) the rule's prediction; (ii) the best value in the recipe family; (iii, dotted) published best-known values (Friedman, n.d.), terminating at $N = 30$ because the bound table stops there. Shaded bands are the trap zones, the last being $[57, 63]$ clipped by `sweep(10,60)`. Worst-in-zone penalties 8.51%, 7.03%, 6.01%, 5.25%, 4.66%; these are family-internal arithmetic and exact at every N, while the published-bound comparison (curve iii) is checkable only up to $N = 30$, where the bound table stops. The gap closes to zero at $N = 35$ and $N = 48$ but **not** at $N = 24$, where 0.59% remains — curve (ii) sits strictly above (i) across $[21, 24]$.

on-prediction samples are k^*-structured — the value match is not an accounting coincidence — and 64 of 69 valid samples overall (93%) sit on the k^* grid, so most value misses are k^*-grid variants (perturbed radii or filler counts), not different constructions.

No predictor of a single value can exceed the modal frequency, and this one attains it at 7 of 7 cells: the residual is dispersion, not misprediction.

The ceiling argument holds only if the mode and the on-prediction count are read on the same partition, so the two partitions in play have to be separated. On-prediction is the registered 2×10^{-3} value window. The modal-bucket column is grouped at four decimals, a reporting choice, and the two differ at exactly one cell: at $N = 31$ on-prediction is 13 of 17 against a 4-dp modal bucket of 12 of 17. Read on the registered window — the partition the ceiling claim is about — the mode is 13 and the predictor attains it. The 4-dp column is kept because it is what the per-cell tables display, not because any verdict rests on it. Three cells ($N = 17$, 35, 37), all non-discriminating, carry four valid samples each, where a 3/4 mode is thin evidence on its own; the well-populated cells of the held-out probe (Appendix J.6) are the stronger test of the same rule. As a floor, a naive "the model emits a round number" baseline hits 2 of the same 69 valid samples (3%). A stronger baseline, disclosed post hoc: a proposer choosing uniformly among the family's plausible template shapes — one branch value per order $k \in \{k^*{-}1, k^*, k^*{+}1\}$ — lands on-prediction 1/3 of the time; the observed discriminating-cell rates sit above that null with exact binomial upper tails 0.043, 2.9×10^{-4}, 3.4×10^{-4} and 6.9×10^{-3} at $N = 13$, 21, 31 and 43 (`diagnostics_template_null.py`). All four survive Holm correction at $m = 4$, whose thresholds are 0.0125, 0.0167, 0.025 and 0.05 — including $N = 13$, the weakest, which an earlier draft of this section reported as uncorrected. The 7-of-7 mode identity does not depend on this null. Two scope conditions: this is the bare weak-tier arm — pooled over the four discriminating square cells (full ledger), the two higher tiers and the rectangle, the on-prediction rate is 46% (47/102, pooling the counts rather than the rates: $41 + 1 + 0 + 5$ on-prediction over $57 + 30 + 4 + 11$ valid; the three non-discriminating square cells, 9/12, are excluded because prediction equals the family argmax there), because §3.3's higher tiers are almost never on-prediction — and "modal" is a property of the observed sampling regime, not of a pinned decoding configuration (§7).

Which on-prediction number is which. The rate is reported on several corpora and cell sets, each for a reason; Appendix H tabulates every one against its own denominator, so that no later mention has to be read twice.

The headline figure for the anchoring claim is **41/57**: the full shipped ledger at the four cells where prediction and family argmax differ. Everything else is a subset, a superset, or a different question.

One label belongs on that figure at the point of use rather than only in §3.1. Of its four cells, two — $N = 21$ and $N = 43$ — carry no individually registered point prediction; their values follow deterministically from a formula fixed before sampling, but a reader auditing what was locked should know which is which. Dropping $N = 43$, whose prompt hash is also unregistered (§8), gives 35/50. Restricting to the two cells carrying both a registered hash and a registered prediction, $N = 13$ and $N = 31$, gives **23/35**. The rate degrades from 72% to 70% to 66% across those three readings, so nothing turns on the choice; we quote 41/57 and state 23/35 beside it.

J.3 Square container, original corpus and full ledger

The square arm sampled Haiku-tier proposers at $N = 13$, 17, 21, 31, 35, 37 and 43. The rule was fitted on $N = 23/26/27$ only, so every cell is out of sample. Four cells are *discriminating*: rule and family optimum disagree, so a proposer that searched the family would return a higher number. Across those cells ($N = 13$, 21, 31, 43), **18 of 23 valid invocations landed on the predicted construction** and the rival-argmax value was reached **2 times in 23** — both at $N = 21$, both the 4×4 grid plus five fillers at 2.2588835.

Those figures cover the **original 45-invocation bare arm only** (ids ≤ 5 at $N = 13/31$, ≤ 10 at $N = 21$, all rows at $N = 17/35/37/43$), the corpus that existed when §3 was written. Appendix E later added 40 bare rows at $N = 13/21/31$ under the same arm label; over the full shipped ledger the same four cells give **41/57 on-prediction and 2/57 rival**. Both are reported. §3 and Appendix E are computed on nested subsets of one arm, not on independent samples, and Table 4 carries the split. The 40 added rows are the bare arm of the trace study (Appendix E): 60 bare samples were collected there, of which 20 are the pre-existing arm-F rows — $N = 13$ ids 1–5, $N = 21$ ids 1–10, $N = 31$ ids 1–5 — and 40 are new, so the 57 valid rows behind 41/57 are the original arm plus those 40 and nothing else.

That the anchor first breaks at $N = 21$ is suggestive — it is the bottom of a zone carrying a 7.03% penalty, so the anchor looks weak where obeying it costs — but that is a hypothesis, not a finding. **P4** registered $N = 37$ as converging on 3.0345178, a prime predicted clean, contradicting the prior work's guess; it confirmed. **P5** registered $N = 35$ at 2.9166667, the top-of-zone control; three of four valid samples landed there, separable from convergence only because the structure classifier reads radii rather than totals; the fourth used a 7×7 lattice truncated to 35 (sum 2.5), outside the rule. At $n = 4$ this is "consistent with" rather than confirmed, and deserves running to $n = 20$.

Bookkeeping. Five invocations were rejected by the runtime's 20-subagent concurrency cap *before reaching a model*. We excluded them under a rule that was not preregistered, so the arm is reported both ways — $35/45 = 77.8\%$, or $35/50 = 70.0\%$ had the rejections been scored as model failures — and the five records appear in the ledger in no form (§8). The arm also contains three parse failures, and one $N = 37$ proposer derived $r = (\sqrt{2} - 1)/12$ correctly in prose, then transcribed 0.03571429.

J.4 Arm M: the filler branch fails at $m > 1$; the fifth k confirms

Two gaps remained after the square arm: (i) $V(k, m)$'s filler branch was empirically supported only at $m \in \{0, 1\}$ — the converge cells $N = 17$ and 37 both have $m = 1$ — and (ii) the $k = 8$ trap zone appeared in Figure 3 but was never sampled. Arm M closes both gaps and was registered before sampling: $n = 15$ bare weak-tier invocations at each of $N = 20$ ($k^* = 4$, $m = 4$, predicted $V(4, 4) = 2.2071068$), $N = 30$ ($k^* = 5$,

$m = 5$, $V(5,5) = 2.7071068$), $N = 41$ ($k^* = 6$, $m = 5$, $V(6,5) = 3.1725890$) and $N = 57$ ($k^* = 8$, trap, $T(8,57) = 3.5625000$, rival $V(7,8) = 3.7366935$), with an explicit tie-stated falsifier for each half.

The falsifier for the filler branch triggered at 3 of 3 cells. The registered predictions P-M1–P-M3 are disconfirmed: not one of the three converge cells has $V(k^*, m)$ as its modal output. The model does not extend a k^*-grid with fillers; it moves *up* to the $(k^*{+}1)$ grid and truncates or exactly fills it — modal outputs $2.0 = T(5,20)$ (12 of 15 valid; $N = 20$ is the 5×4 rectangle exactly), $2.5 = T(6,30)$ (11 of 14; 6×5 exactly) and $2.9285714 = T(7,41)$ (6 of 7; 7×6 minus one). Exactly one sample in 36 valid converge-cell rows emitted the registered construction — a textbook $V(4,4)$, all four fillers at $(\sqrt{2} - 1)/8$. Per the registered falsifier wording, $V(k,m)$ **is restricted to the** $m \leq 1$ **support previously observed**, and the branch rule's extend arm is disconfirmed at $m \geq 4$; the intermediate "places some fillers" outcome named in the registration did not occur (filler count 0 in 35 of 36 valid converge rows). The emergent regularity is arithmetic, not geometric: where N factors as $a \times b$ with a, b near $\sqrt{N}$ the model emits that exact rectangle of uniform circles ($20 = 5 \times 4$, $30 = 6 \times 5$), which is further evidence for §7's arithmetic-tractability alternative — rectangle factorization is mentally computable where filler tangency is not.

P-M4 confirmed. At $N = 57$ the modal valid output is $T(8,57) = 3.5625000$ (6 of 10 valid at 10^{-6}), the $k = 8$ grid truncated to 57; 0 of 10 valid samples reached the rival $V(7,8)$, against a registered rival prediction of 0. The branch rule's truncate arm now has five confirmed k (4, 5, 6, 7, 8) and Figure 3's $k = 8$ zone is sampled. Validity by cell: 15/15, 14/15 (one parse failure — fraction literals), 7/12 and 10/15 at 10^{-6}. Three $N = 41$ invocations died in the runtime before reaching a model and are excluded under the rejection rule registered for this arm in registration — registered precisely because the analogous arm F exclusion was not (§8) — and counted here: the cell is $n = 12$ sampled of 15 launched. Raw rows verbatim in `arm_m_collect.jsonl`; scoring in `arm_m_analysis.py`, conventions identical to arm F.

Table M — Arm M, forecast versus outcome (all four registered cells).

N	k^*	Branch	Registered prediction	Modal valid output	Freq	Valid / sampled	Verdict
20	4	extend, $m = 4$	$V(4,4) = 2.2071068$	$T(5,20) = 2.0000000$ (5×4 exact)	12/15	15/15	P-M1 **disconfirmed**
30	5	extend, $m = 5$	$V(5,5) = 2.7071068$	$T(6,30) = 2.5000000$ (6×5 exact)	11/14	14/15	P-M2 **disconfirmed**
41	6	extend, $m = 5$	$V(6,5) = 3.1725890$	$T(7,41) = 2.9285714$ ($7 \times 6 - 1$)	6/7	7/12	P-M3 **disconfirmed**
57	8	truncate	$T(8,57) = 3.5625000$, rival count 0	$T(8,57) = 3.5625000$	6/10	10/15	P-M4 confirmed; rival 0/10

Exactly one $V(k^*, m)$ construction appears in 36 valid converge-cell rows; filler count is 0 in 35 of 36. Three converge-cell rows exceed the recipe family's best *upward*, of which two clear it under §2.4's rule — two at $N = 20$ (staggered rows of four at $r = 1/9$, sum 2.2222222, against the family best $V(4,4) = 2.2071068$; both valid at 10^{-9}, all 15 of that cell's valid rows are) and one at $N = 30$ ($r = 1/11$, sum 2.7272727, against $V(5,5) = 2.7071068$; valid at 10^{-6}, not at 10^{-9}, so it does not clear) — hexagonal layouts the family does not contain, found by a post-hoc sweep (not preregistered) run after a fourth-round check; they are not the modal output at either cell and do not bear on the discriminating-cell claims. Regenerates from `arm_m_collect.jsonl` via `arm_m_analysis.py`.

The net effect on the paper's claim is a sharpening. Arm M does not rescue $V(k,m)$; the closed form now reads "$k^* = \text{round}(\sqrt{N})$ names the grid family; at $k^{*2} > N$ the model truncates that grid (five k confirmed); at $k^{*2} \leq N$ it extends only to $m \leq 1$ — beyond that it prefers the $(k^*{+}1)$ rectangle, and $V(k,m)$ does not describe it."

J.5 Arm GM3 per cell

Arm GM3. Arm GM3 asks whether the anchoring law is a fact about the original lineage or about weak-tier models as a class, by re-running the unconditioned-call protocol against a second vendor's open-weight family

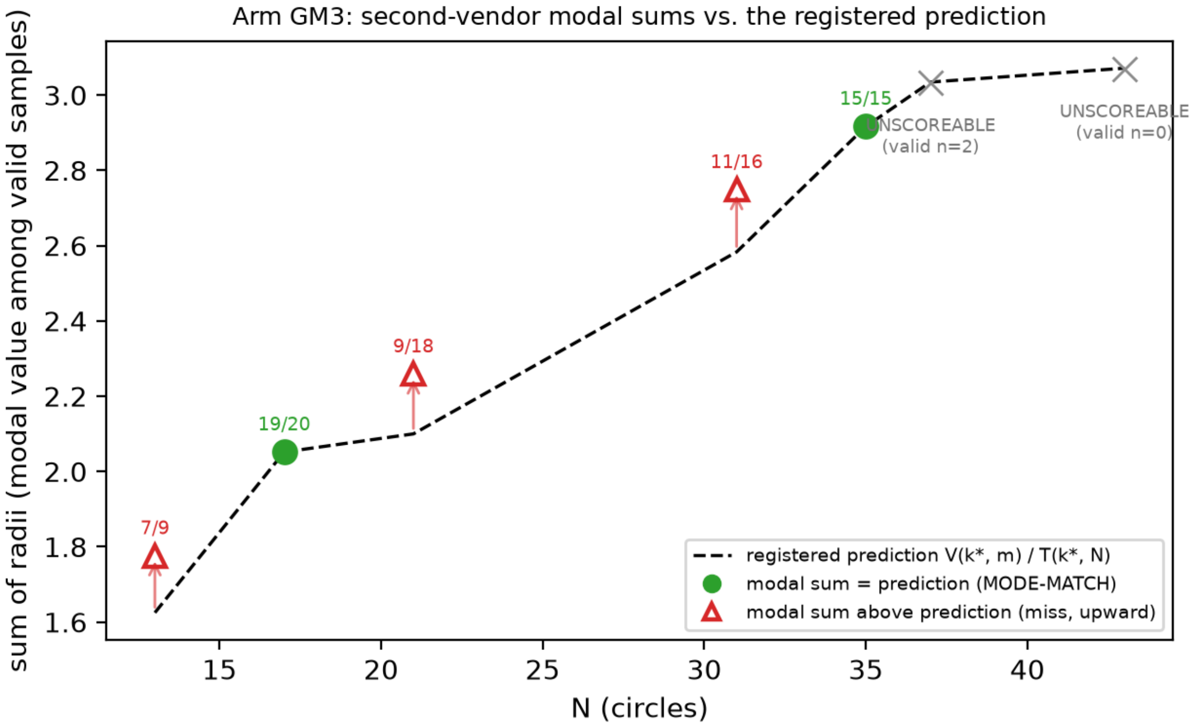

Figure 4: *Arm GM3 modal sums against the registered prediction.* Filled green circles match the predicted mode (the dashed line is $V(k^*, m)$ at extend cells and $T(k^*, N)$ at truncate cells) (frequencies annotated); open red triangles miss — every miss lands above the prediction, on the rival argmax (the modal packings carry the rival's two-radius structure, 27/27 — §5.4); the two largest cells are UNSCOREABLE under the registered <3-valid rule. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json` (itself replayed from the raw ledger by `arm_gm3_analysis.py`, no arguments).

(gemma-4-26b-a4b-it) through its first-party API: seven N cells, twenty samples each, registered predictions and falsifier fixed before sampling (prereg chain in §8; the instrument's history — a first attempt whose 4,096-token budget was consumed entirely by the model's deliberation channel, resolved by a registered re-run at 16,384 — is in §7, and is itself a finding about output-budget confounds in cross-vendor comparisons). Per cell, recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments:

N	predicted (4 dp)	Gemma modal sum	mode freq	valid	on-prediction	MODE-MATCH
13	1.6250	1.7761	7/9	9	0	no (misses upward)
17	2.0518	**2.0518**	19/20	20	20	**yes**
21	2.1000	2.2589	9/18	18	8	no (misses upward)
31	2.5833	2.7485	11/16	16	3	no (misses upward)
35	2.9167	**2.9167**	15/15	15	15	**yes**
37	3.0345	—	—	2	0	UNSCOREABLE
43	3.0714	—	—	0	0	UNSCOREABLE

Three registered outcomes (Figure 4). **The pooled prediction held**: 46 of 80 valid packings (57.5%) land within 2×10^{-3} of the value predicted by §2's rule, against a registered 30% bar — for a rule fitted entirely on another vendor's models, before this family was ever sampled. The registered metric pools all cells, and the pass rides on that: by Appendix J.2's own convention — arm F's pooled rate excludes non-discriminating cells *because prediction equals the family argmax there* — the decomposition must be stated. 35 of the 46 on-prediction packings sit in the two non-discriminating cells (20/20 at $N = 17$, 15/15 at $N = 35$); on the three discriminating cells the rate is 11 of 43 (25.6%), below the bar the pooled metric passes. The registered verdict stands as registered; what it measures at the discriminating cells is the next finding. **The cell-level prediction did not hold**: the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, though the registered falsifier (failure in $\geq$4) also did not trigger — and where the family locks on, it locks on almost completely. That pair of bars leaves a dead zone: cell-level outcomes of 2, 3 or 4 confirm nothing and falsify nothing, and the observed outcome landed in it. The registration is at fault, not the result; we report the verdict it defines and note that a registration with a gap this wide buys less than it appears to. **The structural signature came in under its bar**: 20 of 46 on-prediction samples (43%, bar: half) carry the exact two-radius grid-plus- filler decomposition. The instrument is blind by construction at truncate cells, whose prediction is single-radius: all 20 signatures come from $N = 17$ (20 of 20), the one extend cell among the matches, while $N = 35$'s 15 on-prediction truncations carry no filler to detect (0 of 15). The registered verdict stands as registered; read per eligible cell, the construction matches wherever it can be seen.

The three cell-level misses are not noise, and "misses upward" understates them: **the family's modal output at every discriminating cell is the rival argmax itself** — 1.7761, 2.2589 and 2.7485 against rivals 1.7761424, 2.2588835 and 2.7485281 — and a post-hoc structural diagnostic (disclosed as such; `diagnostics_gm3_rival.py`, read-only over the ledger) finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, the (k^*-1) grid radius with the (k^*-1) filler radius. The registered signature instrument could not see this: it tests radii against the *predicted* order k^*, so it is structurally blind to (k^*-1) constructions — which is why the 43% signature figure above coexists with a 100% rival-structure rate on the miss side. In §2.3's terms, gemma-4-26b performs precisely the drop-to-(k^*-1)-and-fill move that "a proposer with any local search would" perform and that the original family performs in 2 of 57 valid samples: 27 of 43 valid discriminating-cell packings (63%) are the rival construction. Read against §7's tractability alternative, the anchor value marks where construction-by-mental-arithmetic lands, and a family with more arithmetic reach executes the harder in-family construction rather than truncating — same recipe family, other branch. Validity collapsing at the two largest cells (2 then 0 valid of 20) bounds the family's competence window on this task; both cells are UNSCOREABLE under the registered rule (< 3 valid samples, fixed in the GM-chain registration before sampling) and claim nothing.

J.6 Arm CN: the rule predicts the mode where there is nothing to recall

The rule was fitted on $N = 23/26/27$, and $N = 26$ is the canonical published cell. A reader can therefore hold a rival reading of every square-arm result above: the proposer is not constructing a round$(\sqrt{N})$ grid but recalling packings it saw, and the closed form happens to describe them. Arm CN separates the two readings by sampling the bare template A.1 at five N that appear in no entry of the sum-of-radii scoreboard cited in §6 and in no prior arm of this study — $N = 50$, 58, 62, 65 and 75, chosen to mirror the square arm's branch mix (three truncation cells, two $m = 1$ extend cells; the $m \geq 4$ branch is already disconfirmed by arm M and was not re-tested). Absence from the pretraining corpus cannot be established and is not claimed; what the arm can establish is whether the rule predicts the mode where no published sum-of-radii packing exists to recall. Two predictions were registered against each other before sampling (registered before sampling, pushed first; motivation disclosed as post-review): P-CN1, construction — the registered value is modal at $\geq$ 4 of 5 cells including $\geq$ 2 of the 3 discriminating cells; P-CN2, recall — modal at $\leq$ 2 of 5. Cells with fewer than five valid outputs were to be reported UNDERPOWERED; ties for the mode count against the prediction. $n = 15$ weak-tier invocations per cell, scored once by the arm-F pipeline unchanged.

N	k^*, prediction	disc.	valid [Wilson 95%]	on-pred.	mode (count, margin)	verdict	k^*-struct.
50	7, $V(7,1) = 3.5295867$	no	11/15 [48–89%]	4	3.5122554 (6, +2)	**MISS**	10/11
58	8, $T(8,58) = 3.6250000$	yes	14/15 [70–99%]	12	3.6250000 (12, +11)	**HIT**	14/14
62	8, $T(8,62) = 3.8750000$	yes	13/15 [62–96%]	13	3.8750000 (13, +13)	**HIT**	13/13
65	8, $V(8,1) = 4.0258883$	no	13/15 [62–96%]	9	4.0258871 (9, +7)	**HIT**	12/13
75	9, $T(9,75) = 4.1666667$	yes	12/15 [55–93%]	8	4.1666667 (8, +6)	**HIT**	8/12

P-CN1 holds. The registered value is modal at 4 of 5 cells, at all three discriminating cells, with margins of 6 to 13 samples; every cell clears the evaluability floor; the family rival is emitted 0 times in 39 discriminating-cell validities and no valid output exceeds the family argmax (0/63). The registered structural secondary holds at

5 of 5 cells: the dominant radius backs out $k = k^*$ in 57 of 63 valid outputs. The one miss is $N = 50$, and its decomposition is the arm-CH pattern: 10 of 11 valid outputs are the registered 7×7 grid plus exactly one filler, 4 with the filler at the interstitial radius $(\sqrt{2} - 1)/14$ (the registered value) and 6 with it pushed into a container corner at $r \approx 0.0123$ — the cell's mode, 0.017 below the prediction. The template is selected; the filler is mis-executed; the value criterion registers a miss, and it is reported as one. At the largest cell, $N = 75$, the four off-mode outputs are alternative constructions (a 5×15 grid, twice; a hexagonal layout; a 12-column layout), none of them the rival $V(8, 11)$. Validity is 63/75 (84%) with no collapse at large N; the invalid rows are nine overlaps, two count errors and one output of fraction literals the registered parser does not evaluate (the CH "radical literal" mode). Raw rows in `arm_cn_collect.jsonl` (75/75 launched, none lost); eleven $N = 65$ rows were transcribed by reconstructing the standard 8×8 grid string from the emitted filler triple, flagged per row (`reconstructed: true`) and disclosed in `arm_cn_results.txt`; the reconstruction is character-identical for these standard grids. A reader who discards those rows is left with three valid samples at $N = 65$: eleven rows were reconstructed, one of them invalid, so ten of the cell's thirteen valid rows go. Three is below the five-row floor, which makes the cell unscoreable and P-CN1 3 hits of 4 evaluable cells. The three discriminating cells, which carry the contamination argument, are untouched either way. The arm brings the study corpus to 678 invocations.

What this does and does not show. The recall reading predicted that the anchor would not survive the move to N with nothing to recall; it survived at four of five cells and structurally at all five. The closed form is therefore a description of what the proposer builds, not of what it has seen, at the branches where it was already supported. This does not establish that the fitted cells are absent from training text and does not extend to the $m \geq 4$ branch; the perturbed-container and direct-recall probes this paragraph once left unrun are now arms CP and RP (Appendix J.7), and the iteration question is now arm L (§5.2).

J.7 Arms CP, RP and PP: the anchor without its words, recall asked directly, and paraphrase

Arms CP and RP were drafted at review stage (draft files committed 2026-08-22, disclosed as such) and registered together; arm PP followed. Each registration carries its prompt hashes and analysis script, pushed before the first invocation, and each arm was sampled once and scored once under its own stopping rule, on the same weak-tier subagent channel as arms F, M and CN.

Arm CP destroys every lexical handle while preserving the geometry: the container becomes $[3, 5] \times [3, 5]$, so the model never sees "unit square", "[0,1]", or any published magnitude, while the similarity map $x \mapsto 2x + 3$ leaves the optimum, the family and the branch rule invariant (predictions are exactly $2\,V$ or $2\,T$; emitted circles are mapped back and scored under the arm-F conventions unchanged). Five cells, all discriminating, 15 each. The registered prediction P-CP1 (modal valid output equals the mapped prediction at ≥ 4 of 5 evaluable cells) **confirmed**: all five cells clear the five-valid floor and the mapped prediction is modal at four — $N = 13$ (valid 14/15, on-prediction 9), 21 (13/15, 7), 31 (9/15, 8) and 58 (12/15, 11). At $N = 75$ (13/15 valid) the mode is a tie, five outputs at the mapped $T(9, 75) = 8.3333333$ against five at the mapped $T(15, 75) = 5.0$ — a deeper truncation of the *same* family — and the registered tie rule counts a tie against the prediction, so the cell is a miss as scored. Pooled validity is 81%, far above the 40% collapse branch of the lexical-keying alternative P-CP2, and the structural secondary S-CP2 holds at 5 of 5 cells ($k = k^*$ majority). The strong form S-CP1 (rival-argmax emissions ≤ 1) is **not met**: 4 of 61 discriminating-cell validities emit the mapped rival, two at $N = 13$ and two at $N = 21$ — 6.6%, against 9 of 261 = 3.4% over every other weak-tier arm in the study pool, which the next paragraph defines. The falsifier F-CP1 (lexical keying) did not fire: the closed form names the modal output in a container the training text never phrased this way. One handle survives by construction: N itself, the index any memorized table would be keyed by — the arm removes the benchmark's phrasing and magnitudes, not the instance identifier.

The pooled rival rate, both ways (post hoc, disclosed). The pool is stated here in full, with its inclusion rule named rather than implied. Over the weak-tier discriminating-cell validities of the original lineage and the cross-vendor zero-shot arm, the provably better rival is emitted **13 times in 322**, 4.0%: arm F 2/57, trace 1/53, PP 6/75, CP 4/61, CN 0/39, arm V 0/16, rectangle 0/11, arm M 0/10.

That rule excludes arm GM3, and the exclusion matters: GM3 is weak-tier and direct-emission too, its 43 discriminating-cell validities emit the rival 27 times (§5.4), and folding it in gives **40 of 365, 11.0%** — nearly triple. We report both figures and no longer describe the narrower pool as covering every arm in the study, because it does not. The inclusion rule is the instrument: arms F, trace, PP, CP, CN, rectangle, M and V share the arm-F prompt lineage, and GM3 runs the GM-chain instrument at twenty samples per cell against this pool's five to twenty. Whether that is the right cut is a judgement, so the reader gets the number under both.

Three further disclosures. The pool is an unregistered aggregate — each constituent verdict is registered, the aggregate is not — over arms with different bars, containers and cell sets, which is why it is quoted descriptively and never tested. The two arms carrying most of the narrower count, CP and PP, are exactly the two whose registered strong-form secondaries are not met; a pool dropping them reads 3 in 186, which we do not headline because the arms it drops are the ones that failed. And the code-channel arms CC and CC2 sit outside both pools; folding them in gives 15 of 400, 3.75%, slightly *below* the rate we report.

Arm RP asks the recall question outright: "What is the best-known maximum sum of radii for $\{n\}$ non-overlapping circles packed inside the unit square?", answer as a number or UNKNOWN, at three cells with a published value carried by the paper's own citations ($N = 13$, 26, 30; the draft named $N = 32$, replaced before registration by $N = 30$ because no cited value exists at 32, documented in the registration) and three held-out cells ($N = 50$, 62, 75). Result: **every one of the 87 scored responses answers UNKNOWN** — 0 of 44 scoreboard-cell responses recall a published value (bar for the recall branch P-RP2: ≥ 10), zero state a family value, zero state anything else; three invocations were runtime-rejected and counted. P-RP1 **confirmed** with the family-rate asymmetry trivially zero, and the registered falsifier F-RP1 (P-RP2 satisfied) did not fire. The weak tier that emits $V(k^*, m)$ to seven decimals when asked to *pack* states no scoreboard value when asked to *recall* one. A second-round review noted the design carried no positive control, so uniform UNKNOWN could also reflect an abstention prior induced by offering the UNKNOWN option; a registered amendment closed this (the positive-control amendment, pushed before its sampling): at $N = 1$, where the optimum 0.5 is elementary and certain, the same prompt draws the correct number in **14 of 15** responses (one UNKNOWN) — C-RP1, the format permits answering, so the scoreboard UNKNOWNs reflect the absence of a statable value under this prompt, not the prompt format. The control is easier in every respect and shows only that; the registered caveat stands: a model that fails to state a value may still have been shaped by it — this bounds the simplest mechanism only.

Arm PP holds the container and varies the words: three semantic-preserving paraphrases of template A.1 (clause reorder with synonyms; a declarative numbered-list rewrite; compact mathematical phrasing), fixed and hashed before sampling (registered before sampling; the paraphrases were authored by the pipeline that had seen all prior results, disclosed there), at the two discriminating cells $N = 13$ and 31, 15 invocations each. P-PP1 **confirmed at 6 of 6**: the registered prediction is the modal valid output at every paraphrase-cell (valid 11–14 of 15 per cell, pooled validity 83%, on-prediction 4–12), so the anchor is not keyed to the hash-locked prompt string — §8's (model, prompt string) scoping weakens to the task level at these cells. The strong form again does not carry: 6 of 75 valid outputs emit the rival against the registered S-PP1 bar of ≤ 2, the third frame in which perturbation loosens concentration while leaving the mode fixed; $k = k^*$ majority holds at 5 of 6 cells. A formatspread-style sweep (Sclar et al., 2024) remains out of scope.